Panasonic

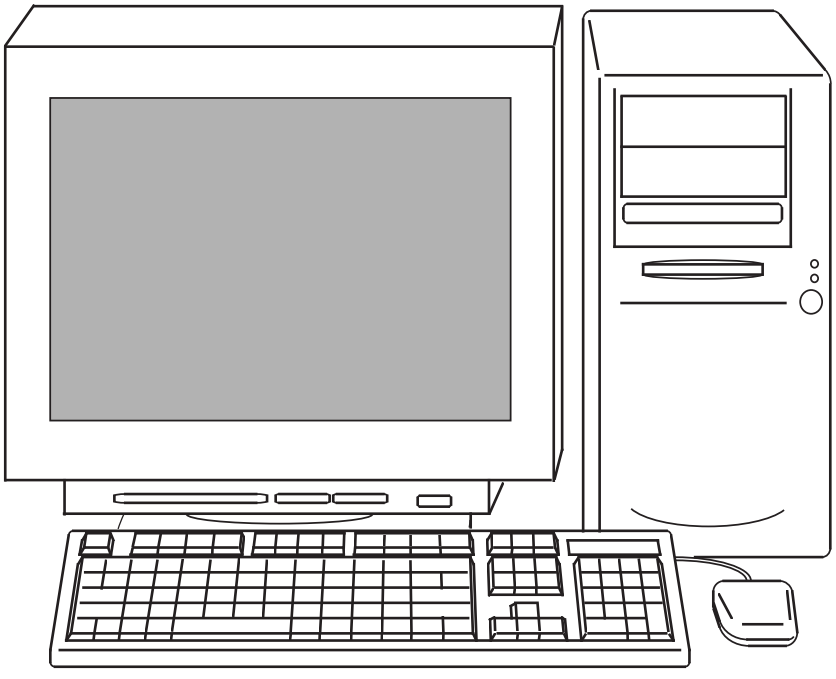
Operating Instructions

Product Changeover and Control

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you for purchasing the “Panasonic product.”

- Read the operating instructions carefully for proper and safe usage.
- Before placing the machine in service, be sure to read “Safety Precautions.”
- Store the operating instructions carefully.

N7201A520E06 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1. OVERVIEW

This chapter explains the overview of PanaCIM-EE / Product Changeover and Control.

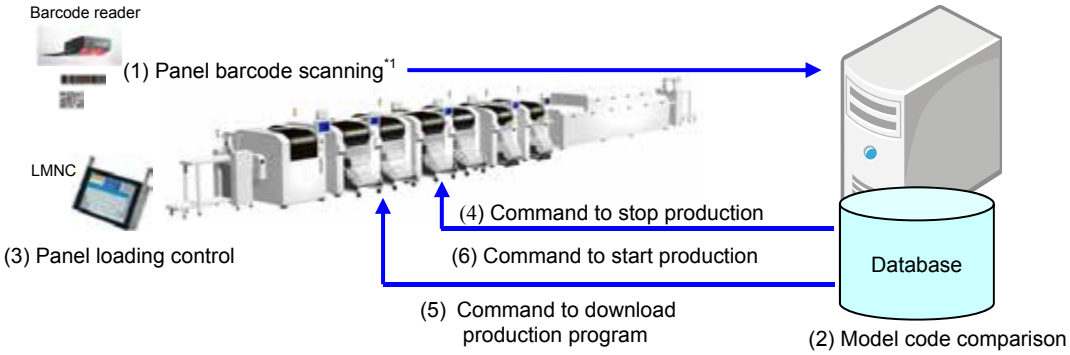
1.1 Overview

PanaCIM-EE (PanaCIM Enterprise Edition: hereinafter referred to as “PanaCIM”) Product Changeover and Control module allows the users to download product programs automatically based on the panel ID, and to adjust the lane width automatically, thus helping to reduce the time required for product changeover.
Product changeover for the NPM series is controlled by hardware ([A] External scanner, [B] Head camera, [C] Plan sheet scanner). (For details, refer to the Specification manual for the NPM series.)*1

*1 AM100,NPM-DX,NPM-D3, NPM-D2, NPM-D ,NPM-TT2,NPM-TT, NPM-W2S,NPM-W2,NPM-W, and NPM.

■ Main functions

(1) Panel barcode definition and scanning
(2) Model code comparison
(3) Panel loading control
(4) Machine control (Command to stop production)
(5) Machine control (Command to download production program)
(6) Machine control (Command to start production)



*1. If a conveyor (connection conveyor) is installed between machines connected in a line, a panel barcode scanner and LMNC may be required for the conveyor.

■ Product Changeover and Control module functions

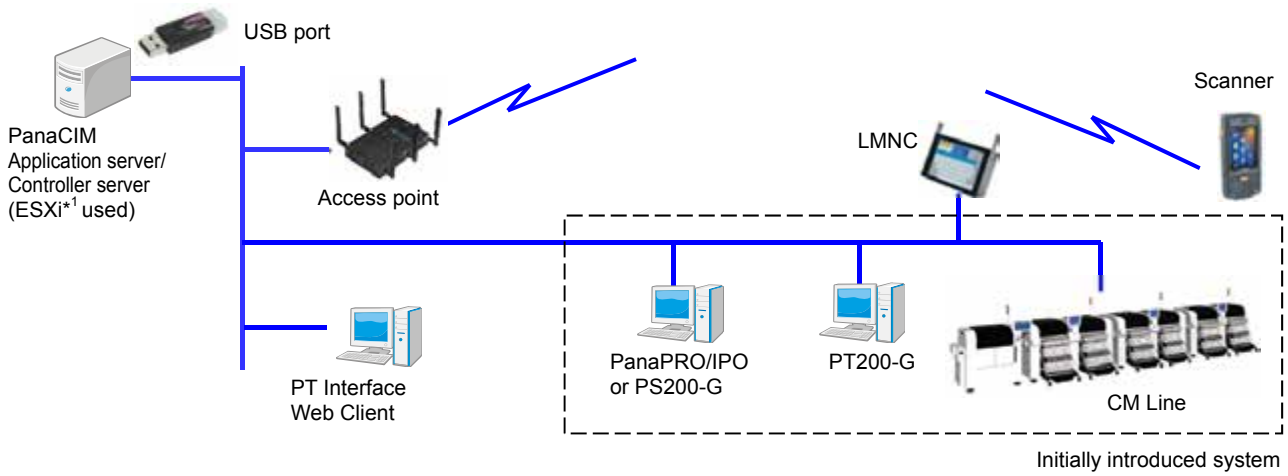
Main functions	Detailed functions		Description
	Large category	Small category	
Barcode Definition	Barcode Definition		Used to define the barcode of the panel used for production.
Panel Barcode Scanning	Barcode Scanning		Used to scan the panel barcodes.
Product Changeover	Model Code Comparison		Used to compare the panel ID model codes.
	Panel Loading		Used to control the panel loading.
	Machine Control		Used to issue the commands to stop production, to download the production program and to start production.

1.2 System Configuration

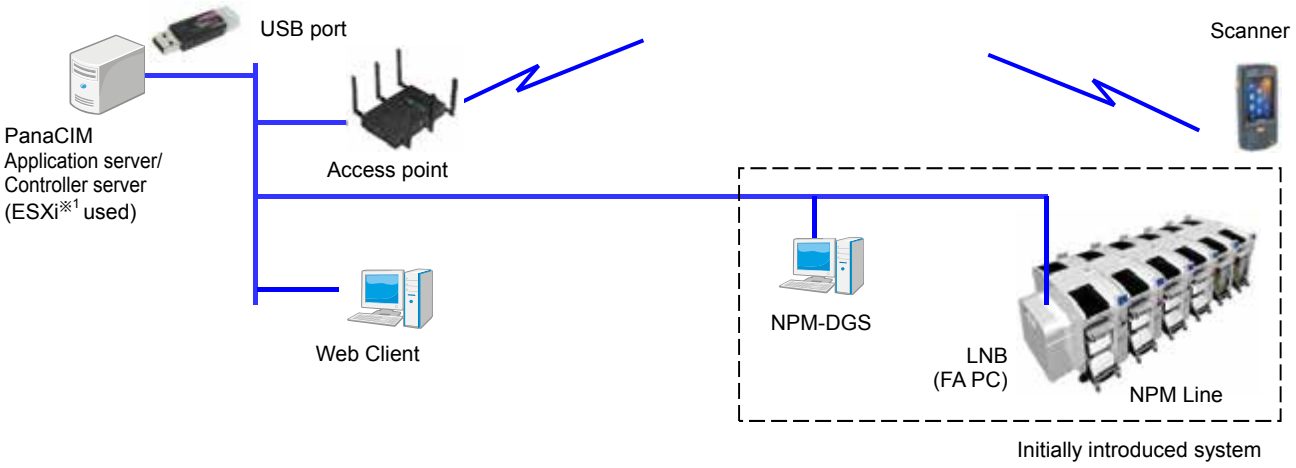
This section shows the schematics of systems with PansCIM for small-scale and large-scale productions.

■ System with PanaCIM (for small-scale production)

CM Line



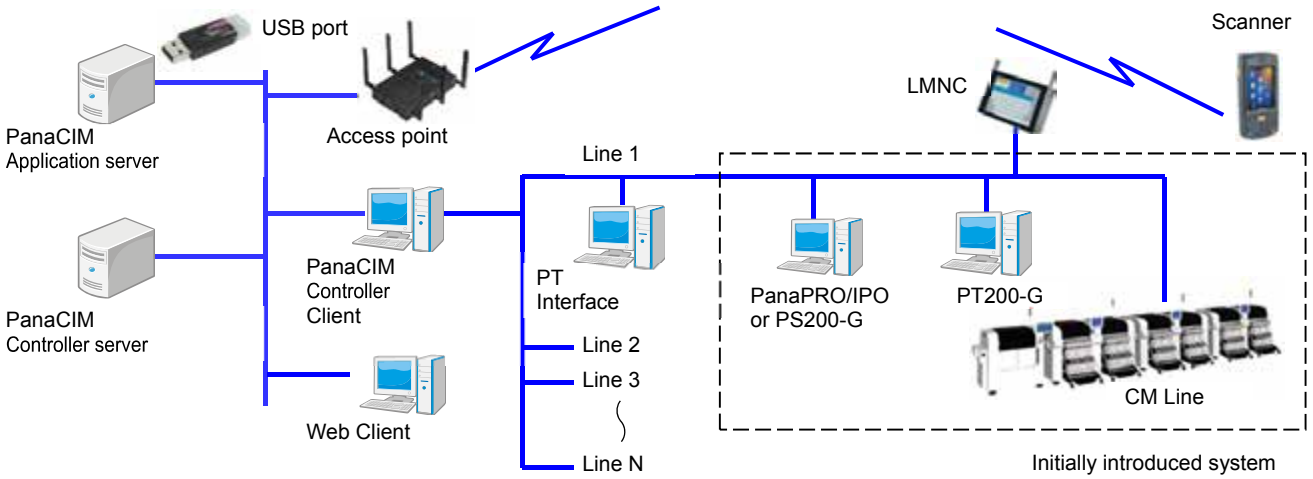
NPM Line*2



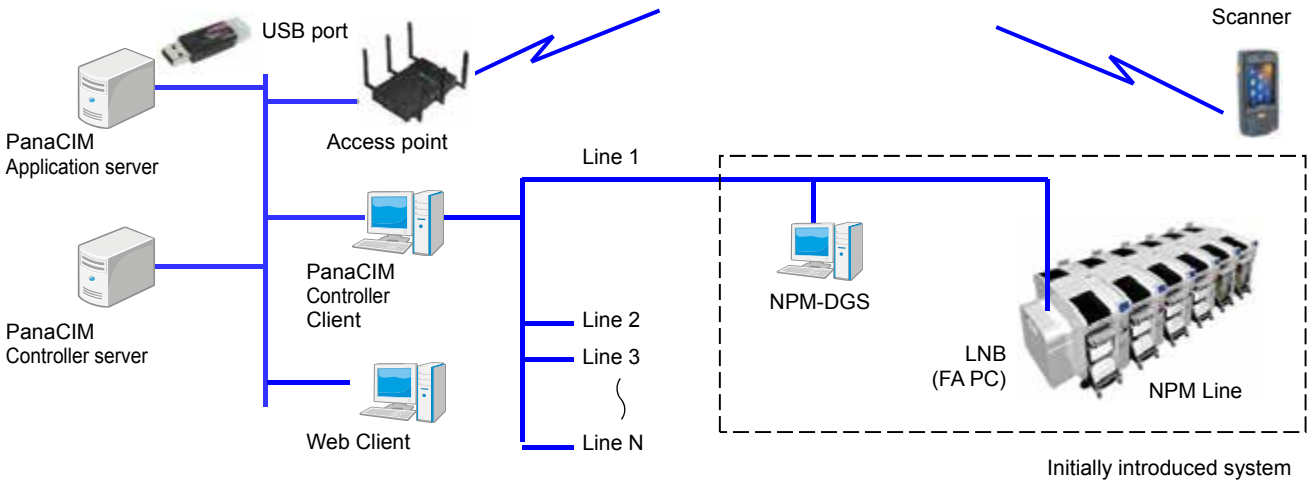
*1. ESXi: A virtualization tool that allows multiple applications to run on a single server.
*2. NPM line can also include the models such as AM100, SPD, SPG and SPV as NPM-related machines.

■ System with PanaCIM (for large-scale production)

CM Line



NPM Line[*1]



*1. NPM line can also include the models such as AM100, SPD, SPG and SPV as NPM-related machines.

1.3 Functional Overview

■ List of Product Changeover and Control functions

Main functions	Detailed functions		Specification
	Large category	Small category	
Barcode Definition/ Scanning	Panel Information	Barcode Format	Supports the following barcode formats:
			1D barcodes: Code 39/Code 128/Code 93/Code 2of5 Interleaved/UPC/EAN/CODEBAR/MSI
			2D barcodes: Data Matrix / QR Code / PDF417
Model Code Comparison	Register Model Code		Links model codes with the product programs before starting production.
	Compare Model Codes		Compares the panel model code scanned in front of the machine and the model code of the current production.
Panel Loading Control	Disable Panel Loading		Prohibits panel loading if the scanned panel model code and the model code of the current production do not agree with each other.
	Enable Panel Loading		Enables panel loading if the scanned panel model code agrees with the model code of the current production. Enables panel loading also after the setup for the machine has been completed.
Error Handling	-		Displays error messages on the LMNC in case of scanning error or if an unregistered model code is scanned.
Machine Control	Command to Stop Production		Commands the machines (in order from the top) to stop production if the scanned panel model code and the model code of the current production do not agree with each other. Commands the next machine to stop production when it has been confirmed that there is no panel on the previous machine.
	Download Product Program		Downloads the product program when production has been stopped.
	Command to Start Production		Commands to start production after the product program is downloaded.
Settings	Control Level	Auto Detect	Prohibits panel loading and stops production if the scanned panel model code and the model code of the current production do not agree with each other.
		Auto Download	Prohibits panel loading, stops production, and downloads the product program if the scanned panel model code and the model code of the current production do not agree with each other.
		Line Managed	Prohibits panel loading, stops production, downloads the product program, and starts production if the scanned panel model code and the model code of the current production do not agree with each other.
	Production Management Data Upload	None	Does not upload (collect) the production management data during product changeover.
		Upload only	Uploads (collects) the production management data during product changeover.
		Upload and Clear	Uploads (collects) the production management data and clears (resets) those in the machine during product changeover.

= MEMO =

2. SETTINGS

This chapter explains the various setting items about Product Changeover and Control.

2.1 Overview

Before starting product changeover and control, the modules for product changeover and control need to be set. This section covers the settings for the CM/DT series machines.

The table below summarizes the PanaCIM controller application and machines setting items used in the product changeover and control modules.

Product Changeover and Control

System Configuration Settings				
	Module Tab			
		Base 1		
			Allow Operator Program Deletes	**Checked**
			Allow Operator Program Uploads	**Uncheck**
			Shared CIM Routes	**Uncheck**
			Enforce Setting of Patterns Per Panel	**Uncheck**
			Machine	**Checked**
			AW	**Uncheck**
			Scanner	**Uncheck**
			Downloads	**Uncheck**
			Select	**Uncheck**
			Adjust	**Checked**
			Enable Base Product Info	**Uncheck**
			Enable product transfer logging	**Uncheck**
			Record Work Order Number	**Uncheck**
			Allow Dynamic Model String	**Uncheck**
		Base 2		
			Enable User Downtime Login	**Checked**
			Maintenance Password	**PANA**
			State Mode	**PanaCIM**
		Product Changeover and Control		
			At Start	**Uncheck**
			At End	**Uncheck**
Route Editor Settings				
	Module Editor Settings			
		Module LM Setup		
			Control Level	**Auto Detect**
				Auto Download
				Line Managed
			Control Mode	**Auto Detect**
				Auto Download
				Line Managed
	Zone Editor Settings			
		General Config		
			Zone Name	**Default**
			Zone Barcode	**Default**
			Panel Queue Size	**3**
		Line Managed Config		
			Control Level	**Auto Detect**
				Auto Download
				Line Managed
			Control Mode	**Auto Detect**
				Auto Download
				Line Managed
		Line Managed Options		

		Management Data	**None**
			Upload Only
			Upload & Clear
		Product Data *1	**All Mix Product**
		Machine Program *1	**Retain after Changeover**
			Delete after Changeover
		Pins/Plate Adjust *1	**Disable**
			Enable
Machine Editor Settings			
	Machine Editor - > Communication Setup		
		Machine Name	**CM402-1**
	Machine Editor - > Communication Setup		
		Port Type	**TCP/IP**
		PT Interface Host PC	**10.140.66.217**
		TCP/IP Port	**8001**
		Cable Name	**None**
	Machine Editor - > Options Setup		
		Management Upload Interval	**5**
		Feeder Mode	**Single**
			Double
		Intelligent Parts Cassette	**Unselect**
		Sends Subslot	**Unselect**
		Enable MIScan Interface	**Check**
		MIScan Name	**MIScan-1**
	Machine Editor - > Communication Setup (For Scanner)		
		Port Type	**TCP/IP**
			Serial
			Terminal Server
			Lantronix TS
		Hostname	**DIGI1**
		TCP/IP Port	**20301**
		Cable Name	**None**
		User Name	**root**
		Password	**dbps**
	Machine Editor - > Supporting Commands		
		Start	**Select**
		Stop	**Select**
		Halt *1	**Select**
		Reset *1	**Select**
		Line Stop *1	**Select**
		Adjust Width *1	**Select**
		Origin Width *1	**Select**
		Supported Deletes *1	**MAI (MAI Program)**
	Machine Editor - > Supporting Data		
		Get Supported	**NC Program**
			Status
		Set supported	**NC Program**
		Management Data	**Production Report**
			Feeder Report
			Nozzle Report
	Machine Editor - > Supporting Data -> Process Order		
		Select Order	**Default**
		Transfer Order	**Default**
		Setup Transfer Files	**MAI (MAI Program)**

		Machine Editor - > Carriage Setup		
			Type	**ZF0 Carriage 1 *(Default)***
				ZF0 Carriage 2 *(Default)*
				ZF2 Carriage 1 *(Default)*
				ZF2 Carriage 2 *(Default)*
			Name	***Default***
			Abbr	***Default***
			Start	***Default***
			End	***Default***
			Shuttle Left	**Uncheck**
			Shuttle Right	**Uncheck**
			Direct Left	**Uncheck**
			Direct Right	**Uncheck**
Additional Equipment Settings				
	LMNC			
		General config		
			Equipment Name	**CM402-LMNC**
			Maintenance Password	**PANA**
		Digital Input/Output Config	**(Digital Output Mask)**	
			SMEMA (Upstream)	**Select**
			SMEMA (Downstream)	**Select**
			Unused	**Unselect**
			Unused	**Unselect**
			Lock #1 (ZF0 Carriage 1)	**Unselect**
			Lock #2 (ZF0 Carriage 2)	**Unselect**
			Lock #3 (ZF1 Carriage 1)	**Unselect**
			Lock #4 (ZF1 Carriage 2)	**Unselect**
			(Digital Input Mask)	
			SMEMA (Upstream)	**Unselect**
			SMEMA (Downstream)	**Unselect**
			Unused	**Unselect**
			Unused	**Unselect**
			Exhaust #1 (ZF0 Carriage 1)	**Unselect**
			Exhaust #2 (ZF0 Carriage 2)	**Unselect**
			Exhaust #3 (ZF1 Carriage 1)	**Unselect**
			Exhaust #4 (ZF1 Carriage 2)	**Unselect**
			Inactive #1 (ZF0 Carriage 1)	**Unselect**
			Inactive #2 (ZF0 Carriage 2)	**Unselect**
			Inactive #3 (ZF1 Carriage 1)	**Unselect**
			Inactive #4 (ZF1 Carriage 2)	**Unselect**
			Machine Empty	**Unselect**
			Zone Empty #1	**Unselect**
			Zone Empty #2	**Unselect**
			Zone Empty #3	**Unselect**
			Zone Empty #4	**Unselect**
			Use Empty Signal	**Select**
		Communication Config		
			Port Type	**LMNC Server**
			Host Name	**CM402-LMNC**
			Port Number	**Default**
			Cable Name	**None**
	Strack			
		General Config		
			Equipment Name	**CM402-Strack**
		Communication Config		
			Port Type	**Muxlink**
			Host Name	**lmnc**
			Port Number	**4**
			Cable Name	**None**
			Parity	

		None
		Odd
		Mark
		Space
	Baud	**9600**
		2400
		4800
		19200
	Bits	**7**
		8

*1: For BM series machines

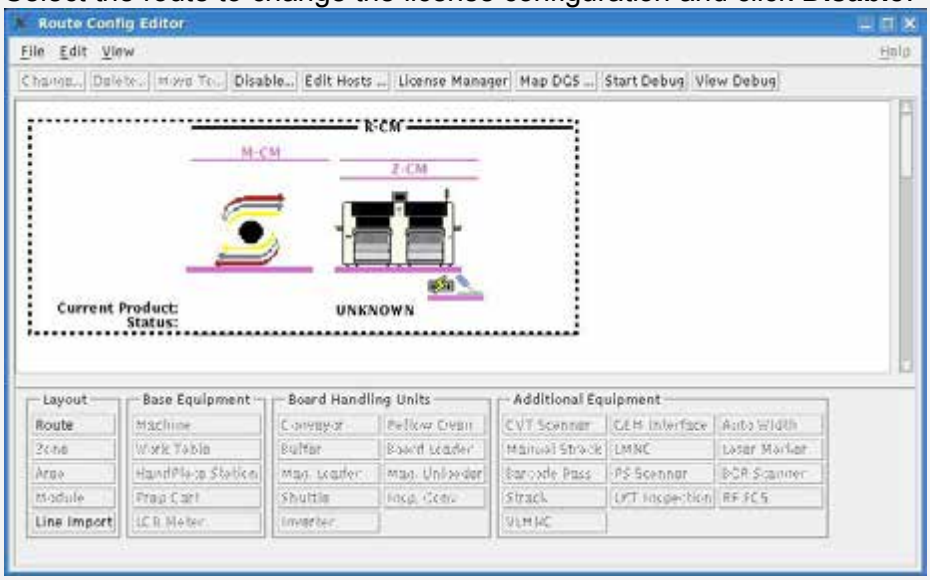
2.2 Settings

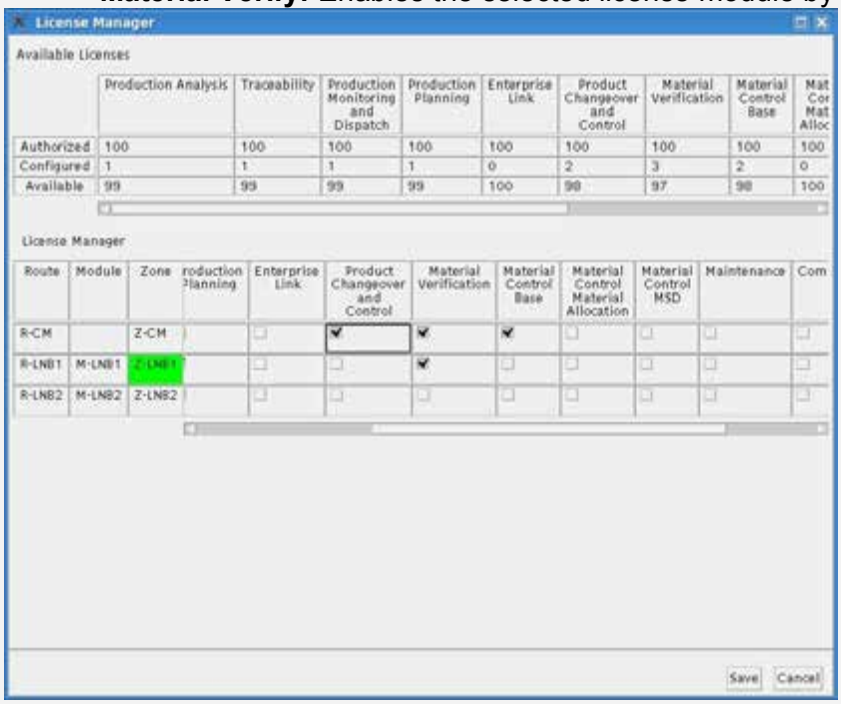
2.2.1 Settings of CIMC

■ Route Editor

● Configure Licenses

Before using the PanaCIM/Product Changeover and Control, the license needs to be enabled via the Route Editor of the PanaCIM controller.

Step	Action
1	Select **Route Editor** icon on the application toolbar of the PanaCIM controller to open the **Route Editor** Application. 
2	Select the route to change the license configuration and click **Disable**.  Clicking **Disable** displays the confirmation dialog. • **Cancel:** Cancels the setting. • **Continue:** Disables the setting. Select **Continue**.
3	From the menu on the toolbar, select **License Manager** to launch the License Manager.

4	Under **Available licenses**, select the license for modules and zones. To use the PanaCIM/Product Changeover and Control, place a check mark in the following checkboxes: • **Auto product change and control:** Enables the selected license zone by zone. • **Material Verify:** Enables the selected license module by module.  **Note:** The lines available in the Route Editor appear in green. To change the configuration of the license, disable the lines by removing the check mark in the box. The PanaCIM/Product Changeover and Control does not support the NPM series machines because the dedicated automatic changeover option is to be used. PanaCIM-based product changeover for the NPM series is offered only as a custom option. Contact us for further information.
5	After configuring the license, click **Save** and exit the License Manager.

Control Mode

The Control Level is the level of functionality purchased by your factory. The Control Mode is dependent on Machine type:

- **CM/DT/NPM series machines (including AM100):** The level is applied to all modules.
- **BM series machines:** The level is chosen for the individual Zones.

Note: Currently, there is no differentiation between license levels, so you should set the Control Level and the Control Mode fields to match when configuring the Route Editor Application.

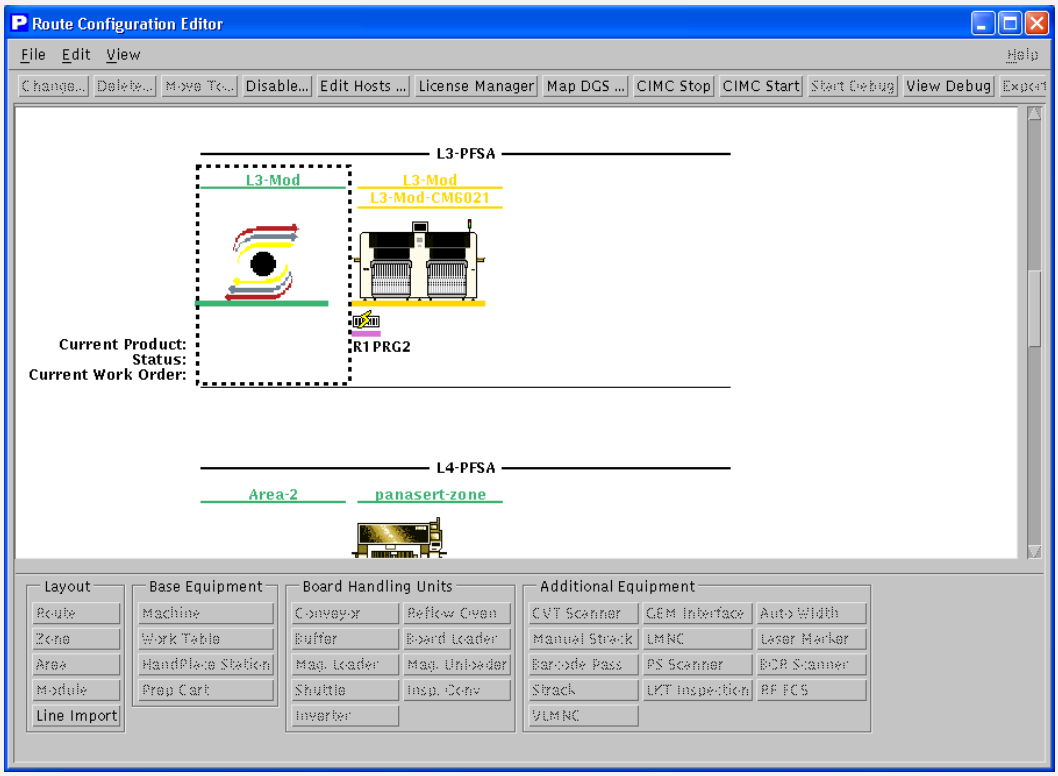
- **Automatic Detection mode:** The operator is informed that a Product Changeover is required and a panel hold is initiated. The operator performs all changeover steps manually and then signals PanaCIM that they are finished. Once finished, the panel hold is released and production will resume

- **Automatic Download:** PanaCIM determines Product Changeover is required and a panel hold is initiated. PanaCIM automatically stops production and selects the program. For BM series machines only.

- **Automatic Changeover mode:** PanaCIM determines Product Changeover is required and a panel hold is initiated. PanaCIM automatically stops production, selects the required program, and restarts production. Little to no operator intervention is required. This is also known as “Line Managed” mode.

- Module Settings - CM/DT Series

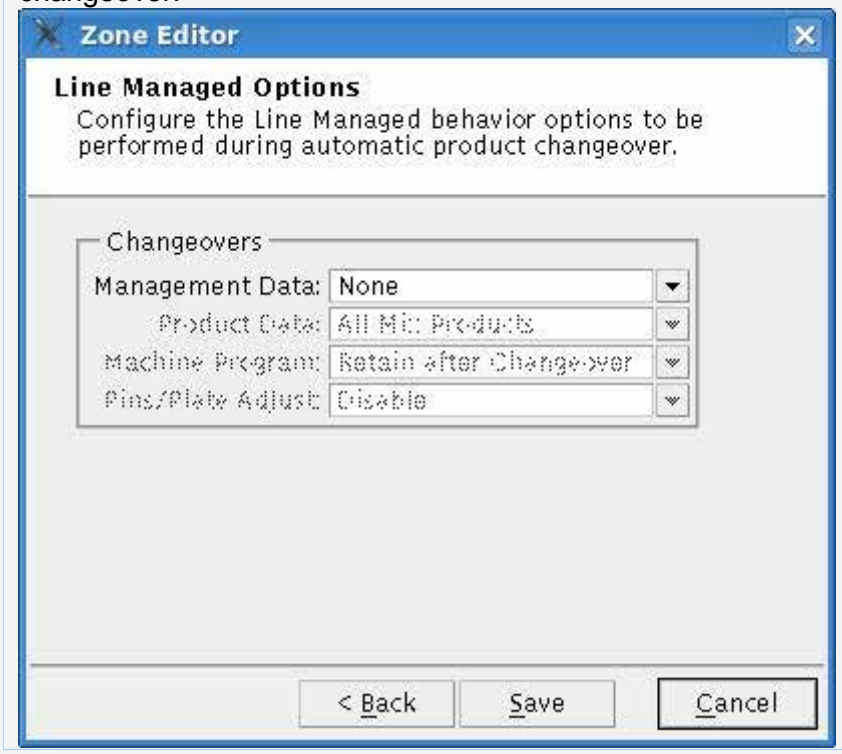
  Refer to the action table below to configure CM/DT series machine settings.

Step	Action
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the Route and click **Disable**.
3	Double click the **Module Object**. 

4	Select your **Control Level** and **Control Mode** from the drop-down list (they should match), and then click **Next**. 
5	Click **Save**.

- Zone Settings - CM/DT and BM Series
  Follow the steps below to set the zone.

Step	Action
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the **Zone** and click **Disable**.
3	Select the zone again.
4	Click **Next** until the **Line Control Setup** page appears.
5	Select the Control Level and Control Mode from the drop-down list and click **Next**. 

6	The line control option page is for specifying the line managed operation (data uploading, etc.) during product changeover. Select the desired option from the drop-down list. **Production management data:** Select the action to be taken for the data managed by the machine or PT200 during product changeover. • **None:** Does not access the production management data during product changeover. • **Upload Only:** Uploads the production management data. • **Upload & Clear:** Uploads the production management data and then clears it for the next production. **Panel data:** For the BM series only. • All Mix Products • Target Product only **Machine program:** For the BM series only. To avoid version mismatch, select **Delete after Changeover**. **Pin/Plate:** For the BM series only. Select **Enable** to display a message by LMNC, prompting to set up the support pins and plate for product changeover. 
7	Click **Save**.

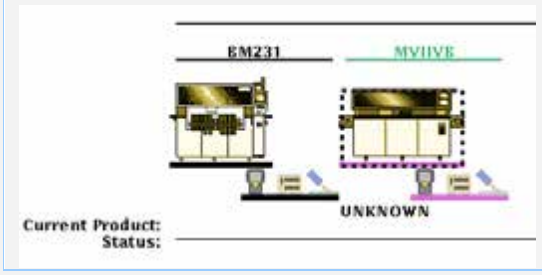
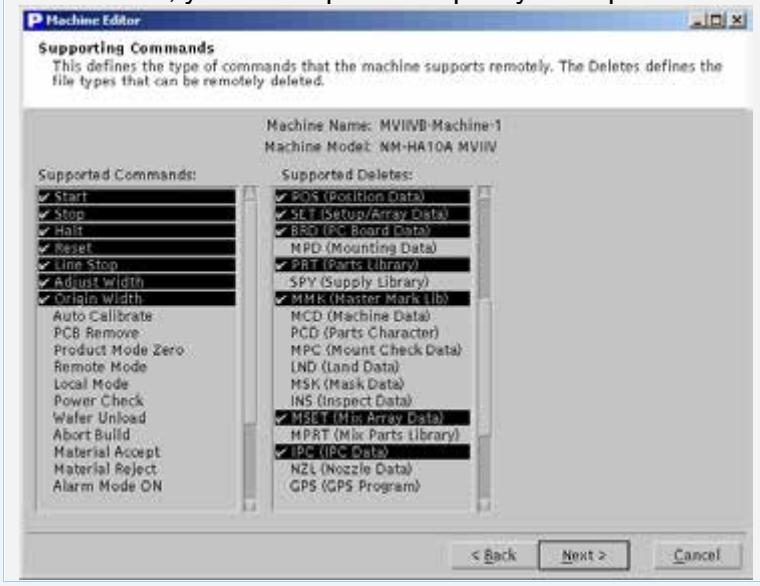
Machine Settings(CM/DT Series)

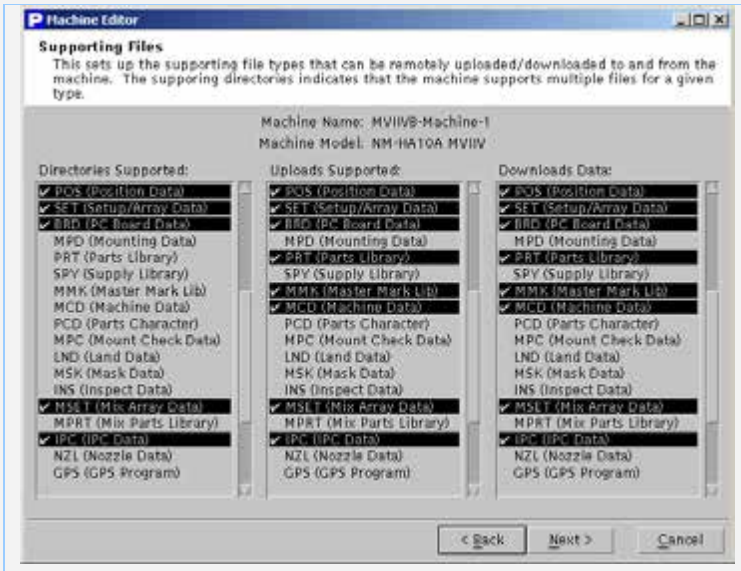
- Machine setting

The machine may require some automatic changes as part of the changeover process.

Refer to the action table below to configure the following:

- Files to download for changeover.
- Commands to the machine that PanaCIM is allowed to use.
- Files that are allowed to be deleted as part of the changeover.
- Programmable controllers other than the machine control that PanaCIM must direct as part of the changeover process.

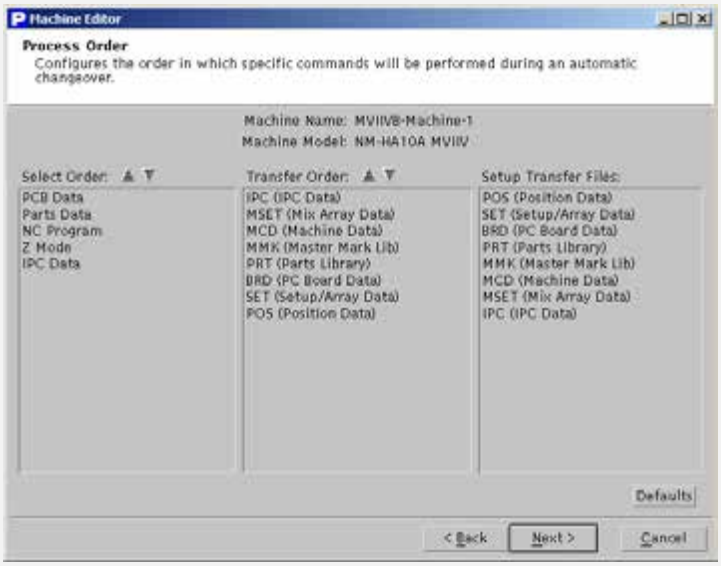
Step	Action
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the **Zone** and click **Disable**.
3	Select the **Machine**. 
4	Select **Next** until the **Supporting Commands** page appears. In order to support an Auto Changeover, the PanaCIM system needs to direct the machine operation. Select the Supported Commands that PanaCIM is allowed to perform to influence the machine operation. It is possible, based on the Line Configured setting in the Zone Setting menus, to set PanaCIM to delete the old program from the machine as part of changeover. In this case, you are required to specify what product files will be deleted. 
5	Click **Next** until you reach the **Supporting Files** page. BM series machines require multiple files to assemble a product. Select the specific files that you desire to be included in a download to the machine and for an upload from the machine.



Click **Next**.

6 It is possible that the order of the information selected at the machine, transferred from PanaCIM, and setup for transfer on PanaCIM, can influence operation during the changeover. The machine type can influence the order the user should select.

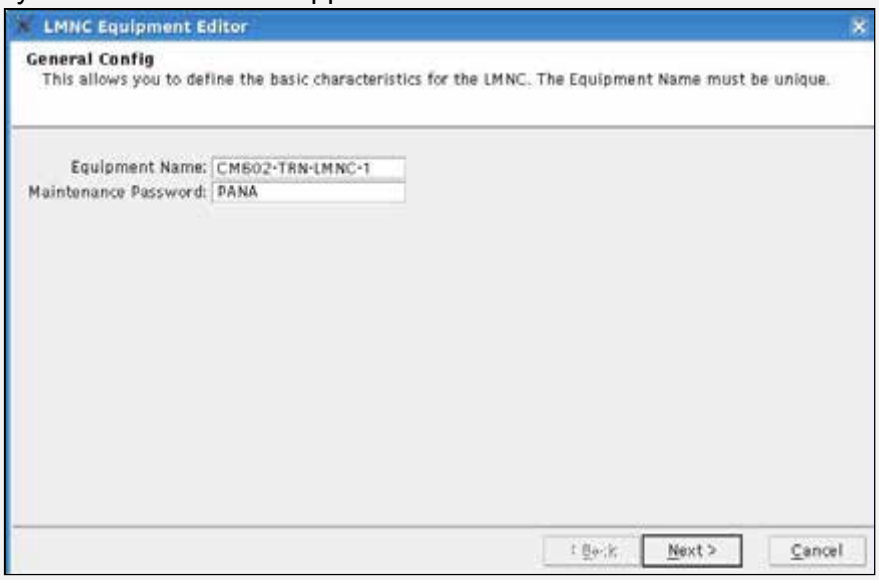
For example, a particular machine may require that the NC program be selected before the Part data. In this case, the operator should move the NC program ahead of the Part Data in the Select Order list.

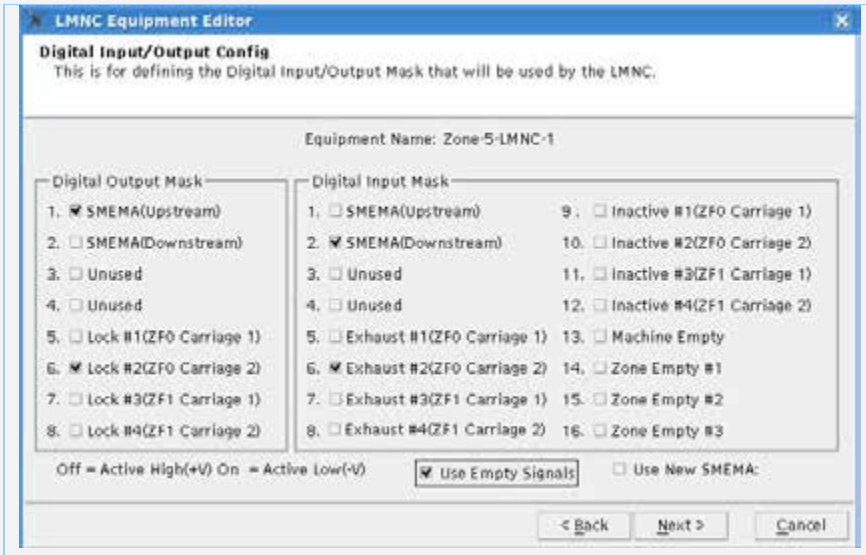


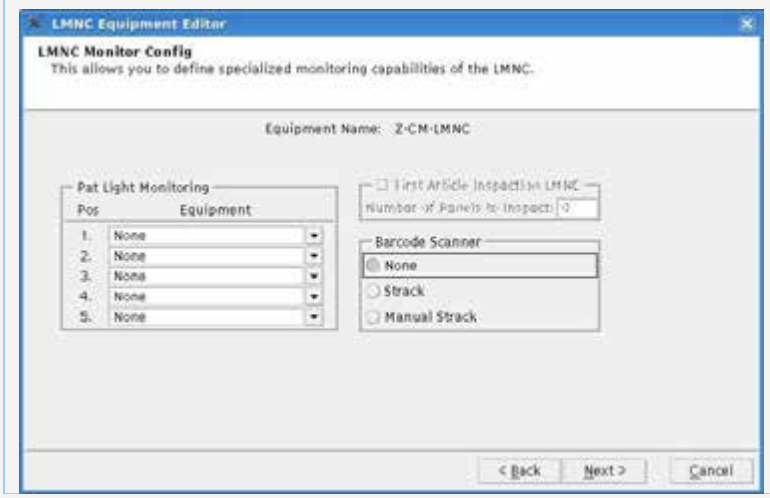
7 Click **Next**, and then **Save**.

Add an LMNC and Panel Scanner to a Zone

In most cases, every Work Zone needs a Line Manager Node Controller and a Panel Scanner. Refer to the action table below to configure an LMNC and Panel Scanner to a Zone.

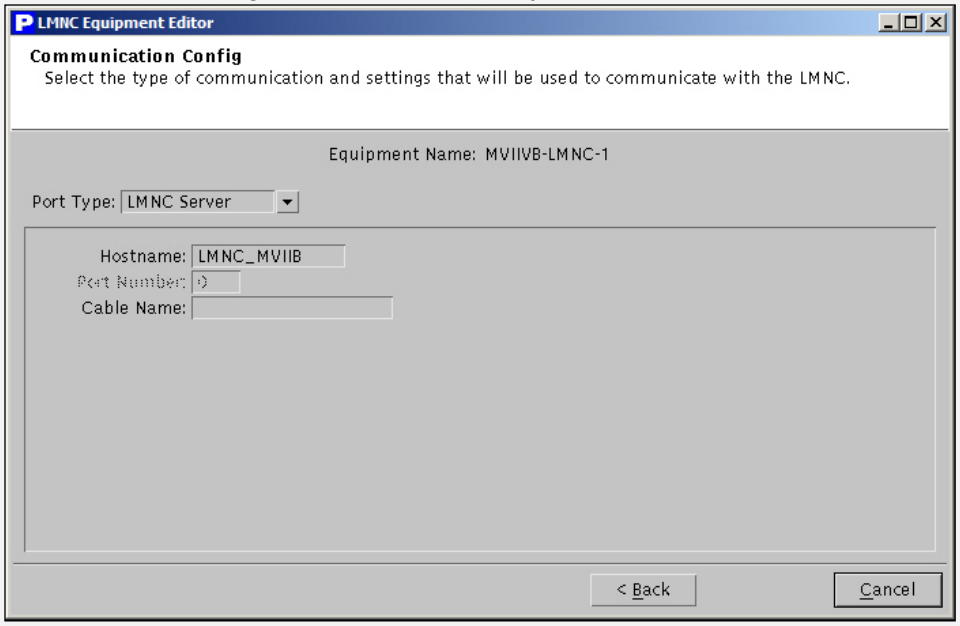
Step	Action
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the **Zone** and click **Disable**.
3	Reselect the **Zone**.
4	Click **LMNC**.
5	Enter the **name** that is assigned to the LMNC and then enter a **password**. This password will be entered at the LMNC display, and it allows the user to perform system override when applicable.  Click **Next**.
6	The Digital Input/Output page determines if each signal is active. *Signals that should be active even if power is removed are normally checked as Low*. The settings required are machine specific but some of the rules that typically hold are as follows: a. Select the Digital Output Masks **1** and **2** for modular machines. This is the only setting for Modular machines. This setting will keep a panel in hold status even if power is removed from the LMNC. b. Check **Use Empty Signals** and then select Digital Input Mask 13 for BM series machines that provide a board empty signal. c. Select Digital Input Masks 5, 6, 9, and 10 for all BM series machines. In addition, select Digital Input Masks 7, 8, 11, and 12 for all BM series machines which have four carriages. d. Click **Next**.





7 The PanaCIM interface to the LMNC is always a LAN connection.

a. Select a **Port Type**. For **LMNC**, always select **LMNC Server**.

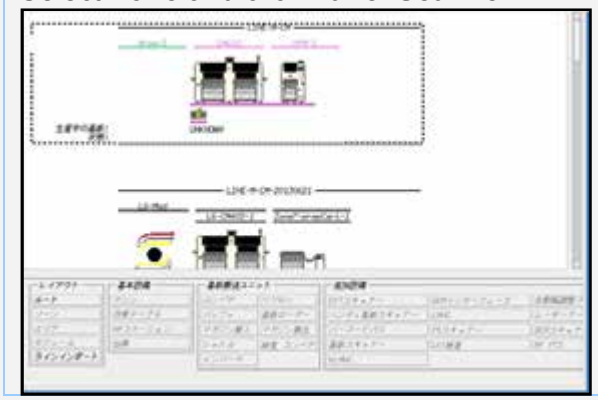


b. Enter a unique **Hostname**. The Hostname must be registered along with the appropriate IP in the **/etc/hosts** file.

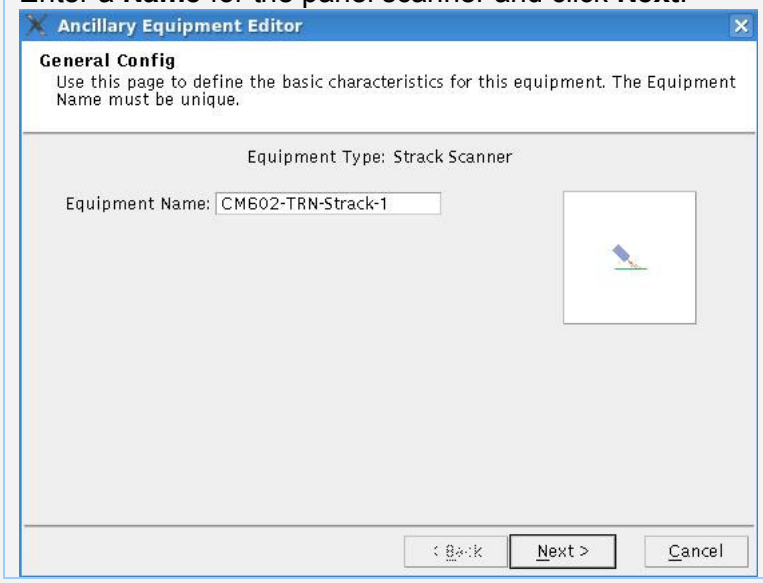
c. Enter a Cable Name.

d. Click **Save**.

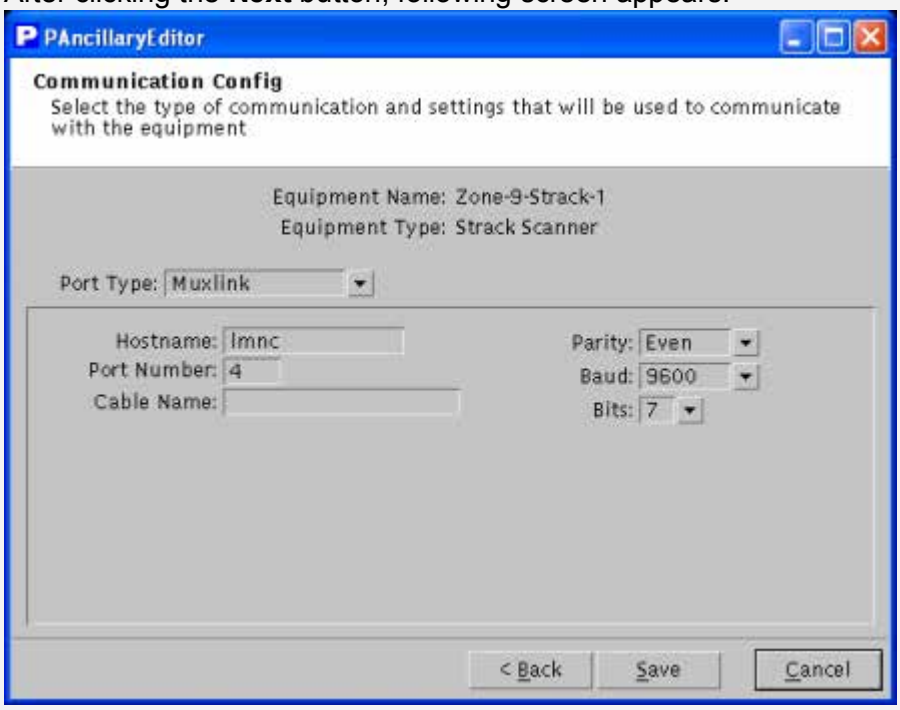
8 Select **Zone** and click **Panel Scanner**.



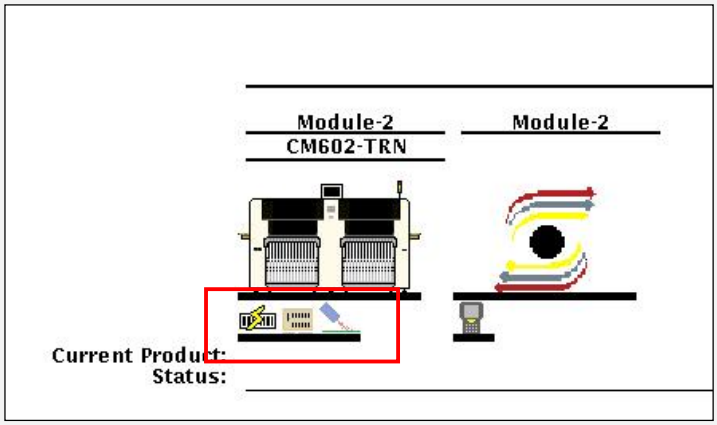
9 Enter a **Name** for the panel scanner and click **Next**.



10 After clicking the **Next** button, following screen appears:



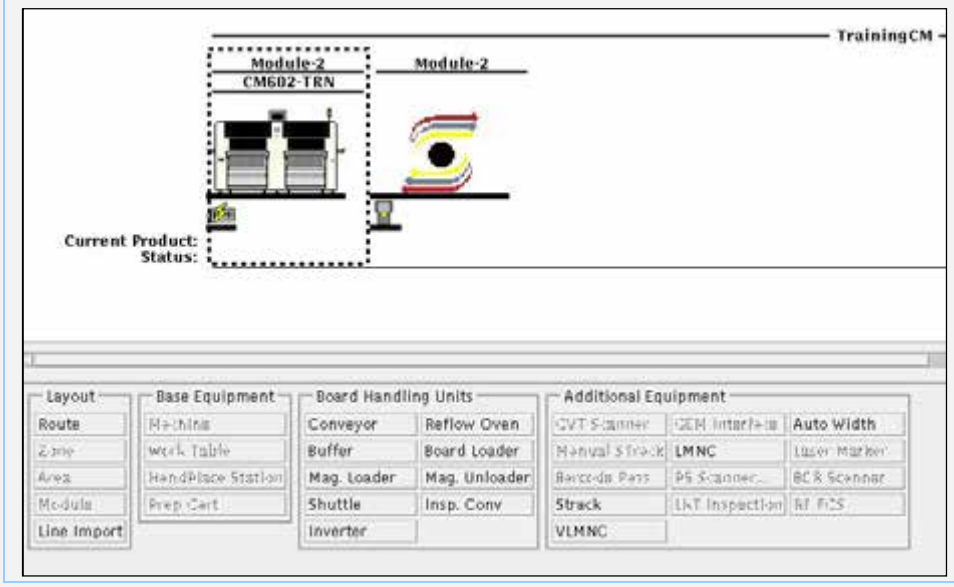
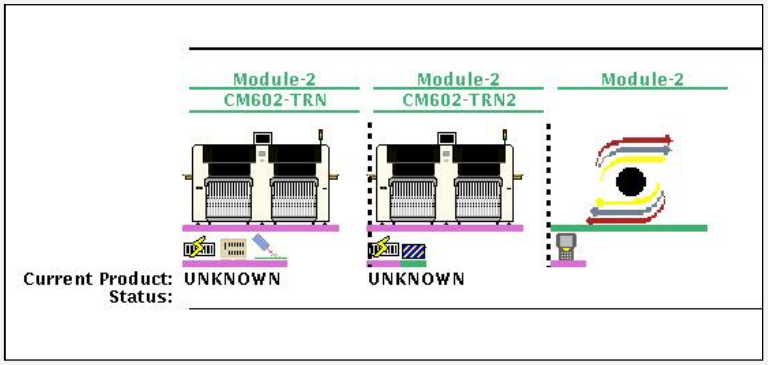
Select the type of communication settings that will be used to communicate with the equipment from the **Parity**, **Baud** and **Bits** drop-down list respectively:

	Default settings shown in the above screen are sufficient for most purposes.
11	Click **Save**. The LMNC and panel barcode reader will be shown below the Zone. 

Add a VLMNC to a Zone

All Zones containing CM/DT series machines (other than the first in the Work Zone) must have a Virtual Line Manager Node Controller (VLMNC).

Refer to the action table below to configure a VLMNC to a Zone.

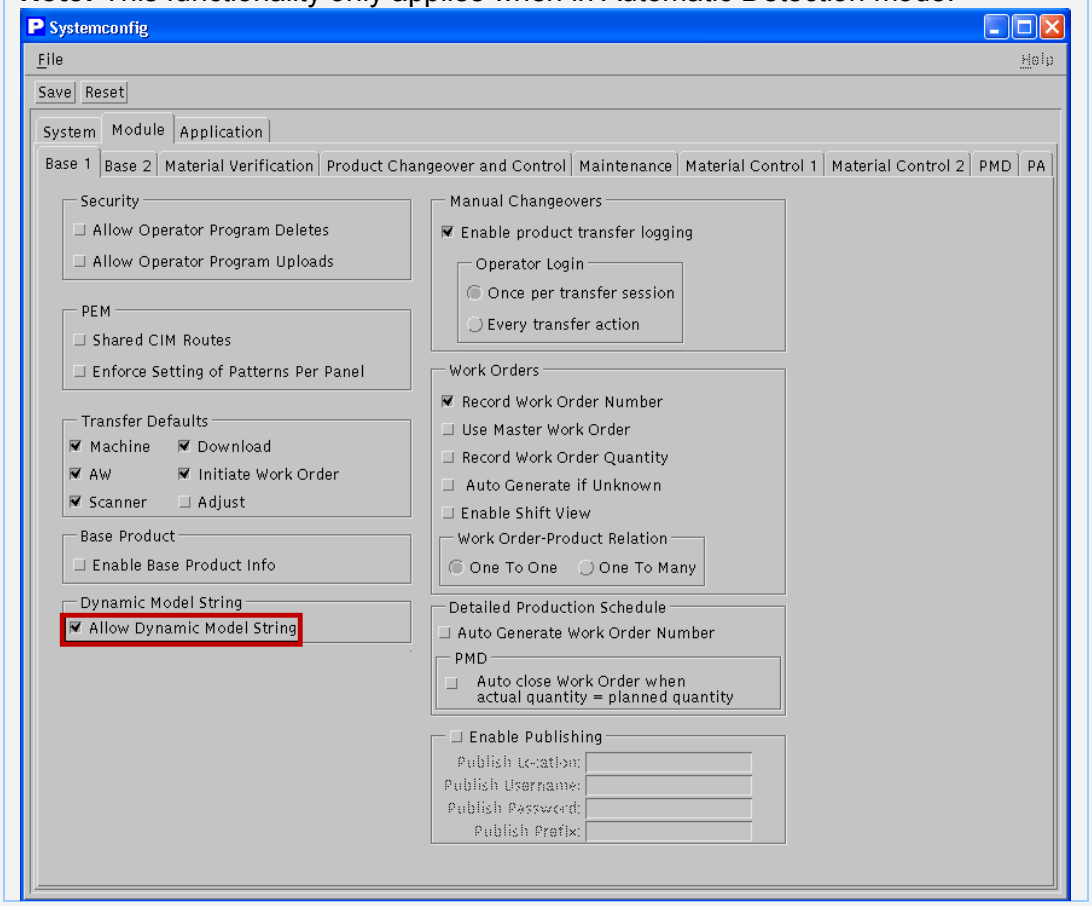
Step	Action
1	Open the **Route Editor** Application from the PanaCIM Main Menu.
2	Select the **zone** that will include the VLMNC and click **Disable**.
3	Reselect the **zone**.
4	Click **VLMNC**. 
5	The VLMNC icon will be displayed in the Zone below the Machine. It indicates that the control of the machine is dependent on the LMNC on the previous machine. 
6	Click **Save**.

■ System Config

● System Configuration Settings

Refer to the action table below to configure the following settings specific to Product Changeover and Control using the System Configuration Application:

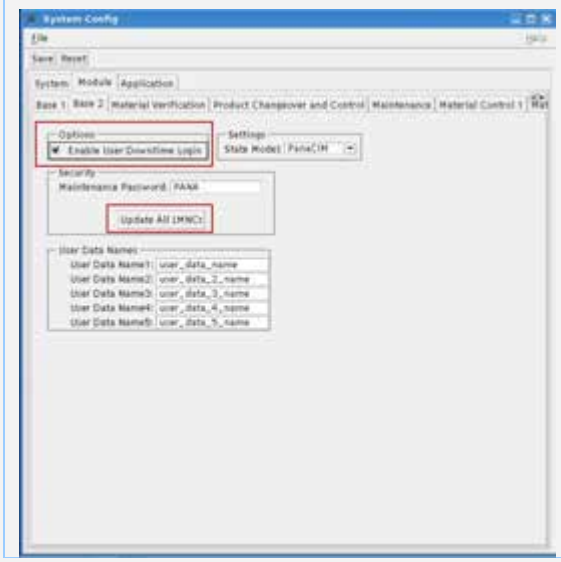
- Provide operators the ability to change the Model Code during Product Transfer.
  * This function is not supported.
- Provide operators the ability to select a reason why a machine has been taken out of production.
  * This function is not supported.
- Update all LMNC passwords at one time.
- Require operators to login when he starts or ends a panel hold at the LMNC.
  * This function is not supported.
- Add attributes operators can choose from when holding a panel.
  * This function is not supported.
- Set machine downtime states.
  * This function is not supported.
- Configure Barcode Expressions.

Step	Action
1	Open the **System Configuration** Application from the PanaCIM Main Menu.
2	Provide operators the ability to change the Model Code during Product Transfer. a. Select the **Module** tab. b. Select the **Base 1** tab. c. Select the **Allow Dynamic Model String** checkbox. **Note:** This functionality only applies when in Automatic Detection mode. 
3	• Provide operators the ability to select a reason why a machine has been

taken out of production using the Route Status Application.

- Update all LMNC passwords at one time.

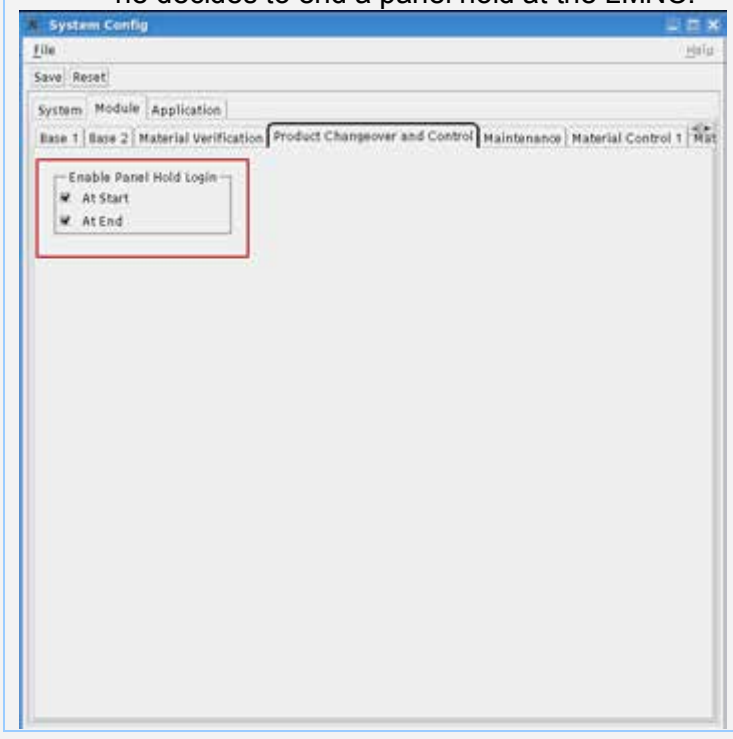
a. Select the **Module** tab.
b. Select the **Base 2** tab.
c. Select the **Enable User Downtime Login** checkbox.
d. Enter the **Maintenance Password** and select **Update All LMNCs**.



4

Require operators to login when he starts or ends a panel hold (stops loading panels by LMNC during panel loading) at the LMNC.

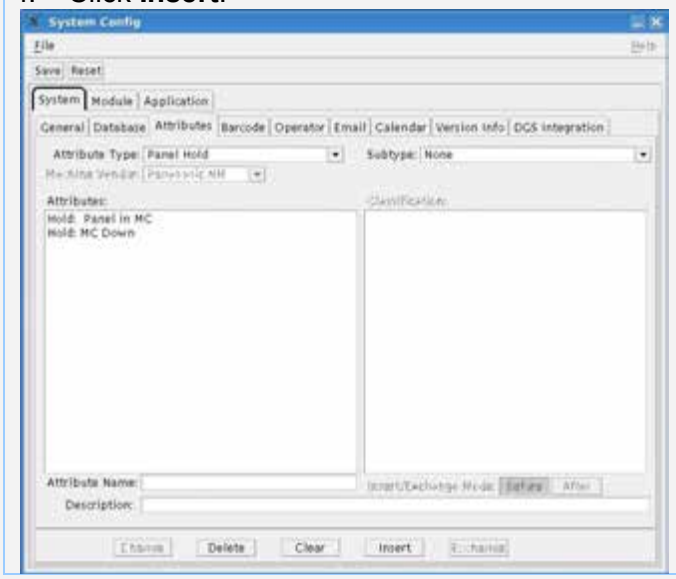
a. Select the **Module** tab.
b. Select the **Product Changeover and Control** tab.
   - Check Enable Panel Hold Login **At Start** to require operators to login when he starts a panel hold at the LMNC.
   - Check Enable Panel Hold Login **At End** to require operators to login when he decides to end a panel hold at the LMNC.



5

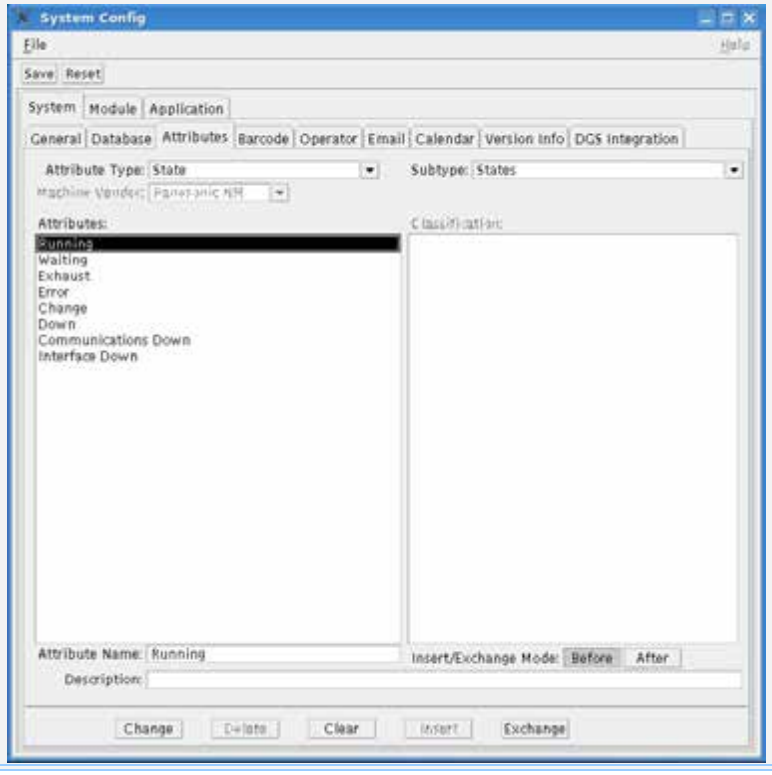
Add attributes operators can choose from when holding a panel.

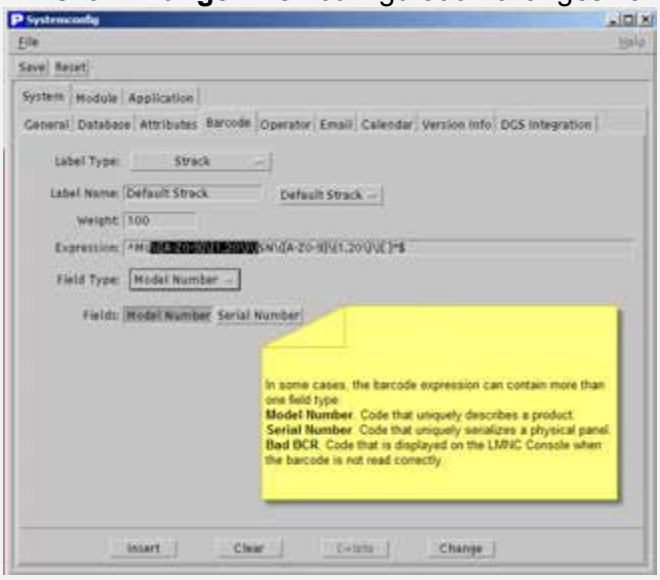
a. Select the **System** tab.
b. Select the **Attributes** tab.
c. Select **Panel Hold** from the Attribute Type drop-down list.
d. Enter an **Attribute Name**.
e. Enter a **Description**.
f. Click **Insert**.



6 Modify machine downtime states that display in the Route Status Application.

a. Select the **Attributes** tab from the **System** tab.
b. Select **State** from the Attribute Type drop-down list.
c. Select **Scheduled** or **Unscheduled** as the Subtype.
d. Enter an **Attribute Name**.
e. Enter a **Description**.
f. Click **Insert**.



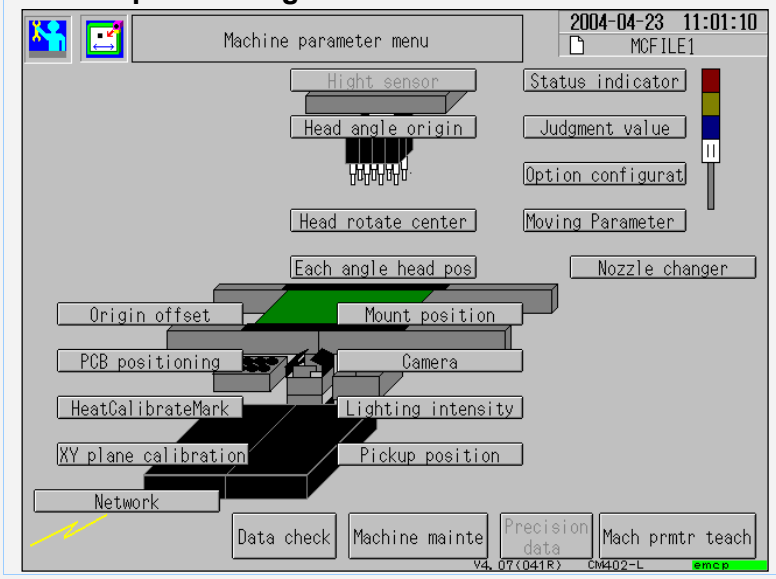
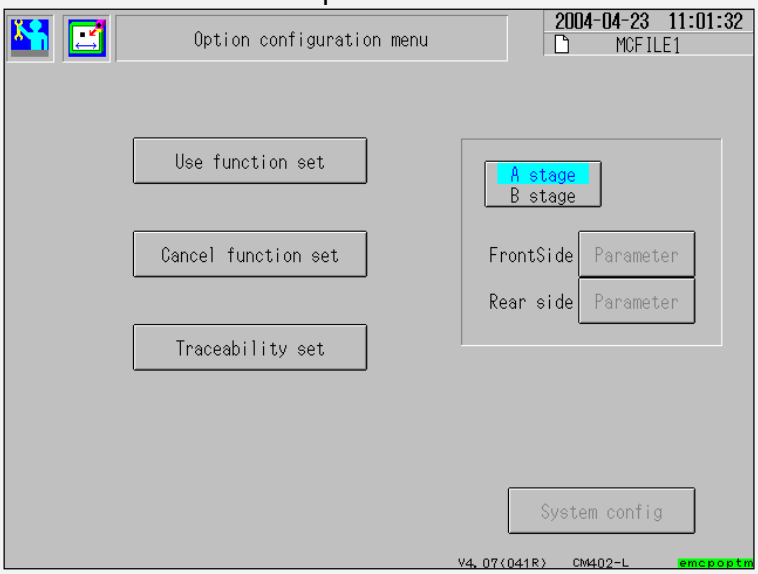
7	Configure how Product Changeover and Control information is formatted for import into PanaCIM. **Note:** Refer to the PanaCIM Base Manual - 7. APPENDIX for detailed instructions on how to set up barcode expressions. a. Select the **Barcode** tab from the **System** tab. b. Select **Panel** as the Label Type. c. Enter a **Label Name** of your choice. This name can be selected in the future to review or modify the expression. d. Enter the **Weight** factor. The weight factor is used when more than one expression will apply to the data. The expression with the higher weight factor will apply. e. Enter the **Expression** (see the Functional Specification document for Regular expression descriptions). f. Select the **Model Number** term from the expression. The selected term will be reverse highlighted. g. From the **Field Type** list choose Model Number. h. **Repeat** this process for the Serial number and the Bad BCR values as needed. i. Click **Change** when configuration changes have been completed. 
8	Click **Save** when configuration changes have been completed.

2.2.2 Settings of Machine

■ CM/DT Series

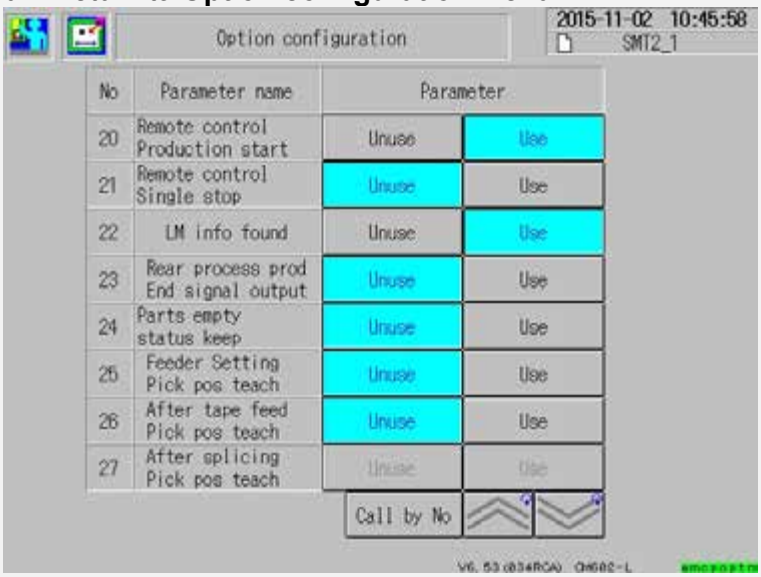
● Setting Machine Options

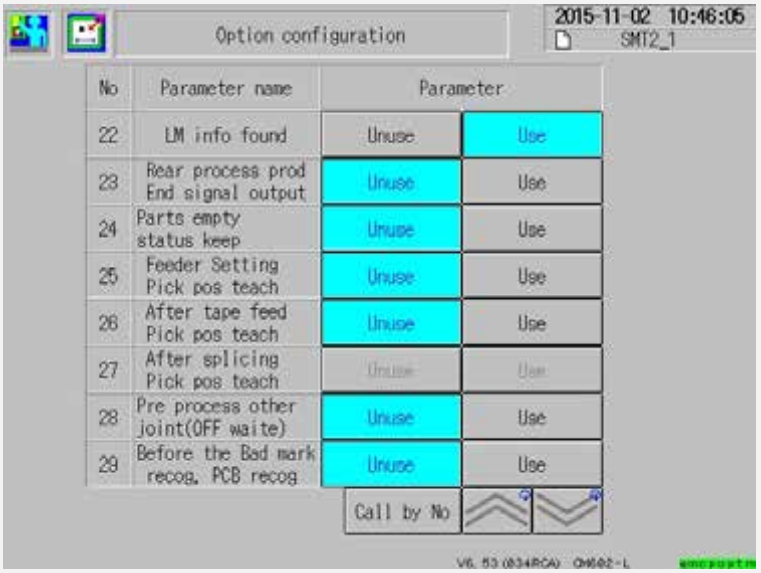
Refer to the action table below to configure CM/DT series machines to be able to operate with Product Changeover and Control *(does not require a Panasonic Service disk)*.

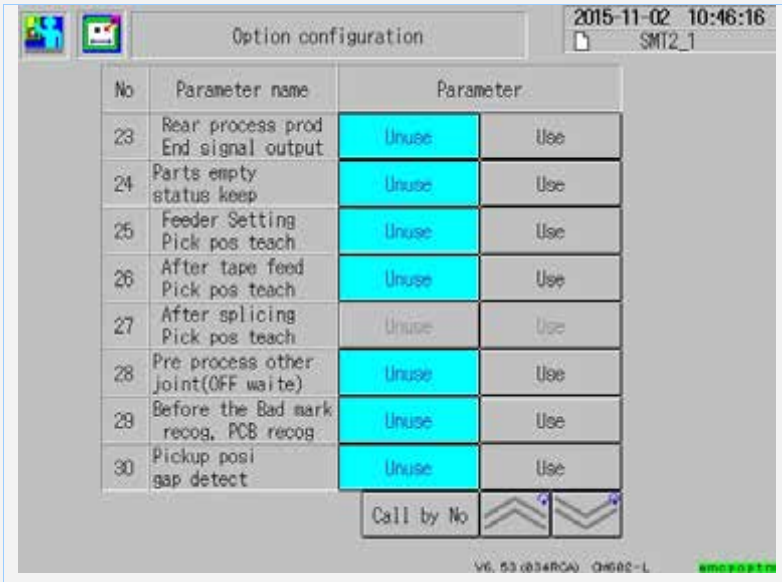
Step	Action
1	Start with the machine in the main menu with the Engineer functions available. Select **Machine parameter**. 
2	Select **Option configuration**. 
3	If the product Changeover and Control is to be run in *Line Managed Mode*, the machine must be set for *remote control operation*. Select **Use function set** and then follow the below steps.  a. Set **Remote control production start** to **Use** b. Set **LM info Found** to **Use**

c. **Set Rear Process Production End Signal output** to **Un use** (This must be turned off or the machine will not respond to PanaCIM commands to stop production) **Note:** The Rear Process Production End Signal output option may be in the Cancel Function menu based on Machine type and firmware level.

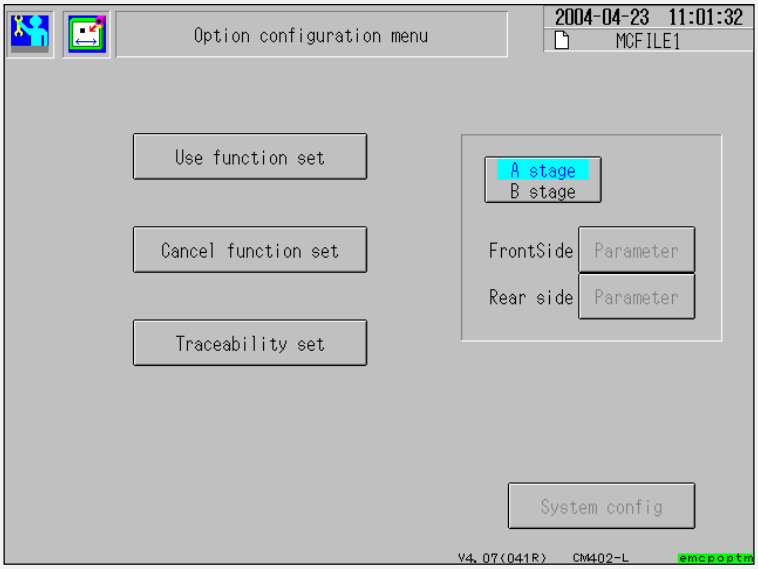
d. Return to **Option configuration** menu.





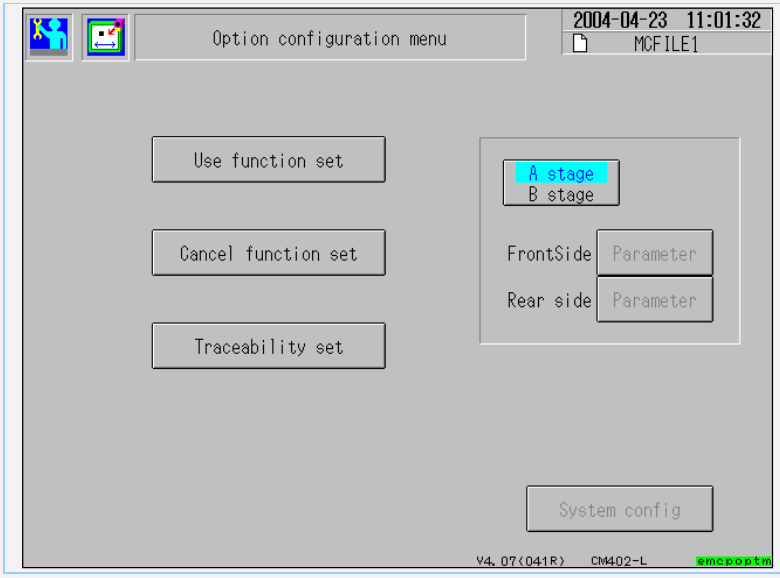


4 If the Product Changeover and Control is to be run in *Auto Detect mode*, the machine must be set for *local operation*. Select **Use function set** and then follow the below steps.



a. Set Remote control production start to **Un Use**.
b. Return to **Option configuration** menu.

5 Select **Traceability set**.



6 Set Pre-pro Barcode Comm and Post-pro Barcode Comm settings according to the position of the machine within the Module.

LM PRE/POST Barcode Communication		
Machine Position	Pre-pro Barcode Comm	Post-pro Barcode Comm
First Machine	USE	USE
Middle Machine	USE	USE
Last Machine	USE	Unuse
Single Machine	Unuse	Unuse

7 Return to the Main Machine Menu when completed.

3. FUNCTION - BASIC OPERATIONS

This chapter explains the basic operations of Product Changeover and Control.

3.1 Overview

After all of the required configurations have been made, you may begin to perform Product Changeover and Control operations.

In this chapter:

- The Applications used to perform basic Product Changeover and Control operations are defined.
- These basic operations are described in detail and are laid out for you in step-action tables.
- The basic operations described include example images to assist you in performing these operations.

Refer to the below table for a brief overview what operations can be performed by using a specific PanaCIM Application.

Application	Overview
Product Setup	In order for Product Changeover and Control to use a product, you must add information to the Product Setup after the product has been imported into PanaCIM.
Route Status	Use the Route Status Application to perform various Work Zones activities during Product Changeover and Control operations such as: • View Events • Configure Machine Downtime • Hold Panel(s)
Changeover Operations	You may have to perform changeover activities if an issue is determined during changeover monitoring. • **During Automatic Changeover on CM/DT Series Machines:** If PanaCIM receives a machine event stating that the feeder and nozzle setup is not correct, you will need to correct any problems and then manually initiate production on the machine. • **During Automatic Changeover on BM Series Machines:** If PanaCIM determines that machine setups in the Work Zone require manual feeder changes, the LMNC display will instruct you to make them.

3.2 Settings

For Product Changeover and Control to use a product:

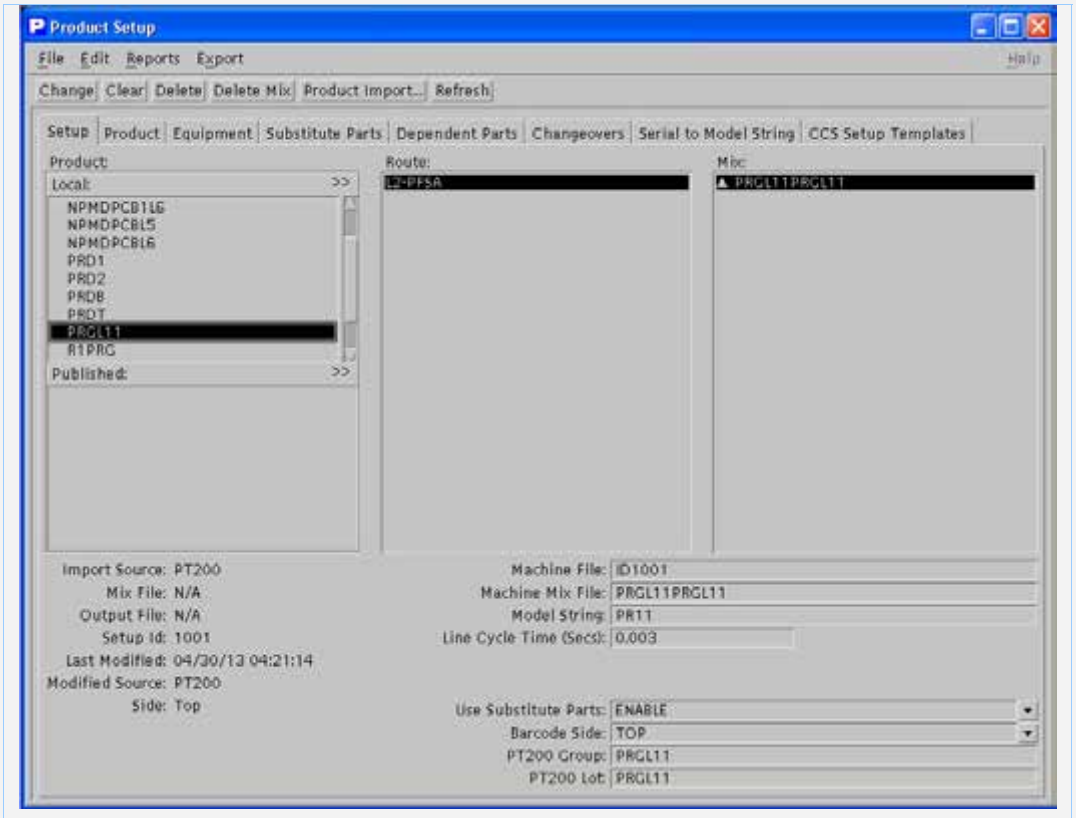
- You must add information to the Product Setup after the product has been imported into PanaCIM.
- After a product has been successfully downloaded from PanaPRO, DGS, or uploaded using Product Import, the product must be setup in the Product Setup.

3.2.1 Settings of CIMC

■ Product Setup

Refer to the action table below to add information to the Product Setup Application.

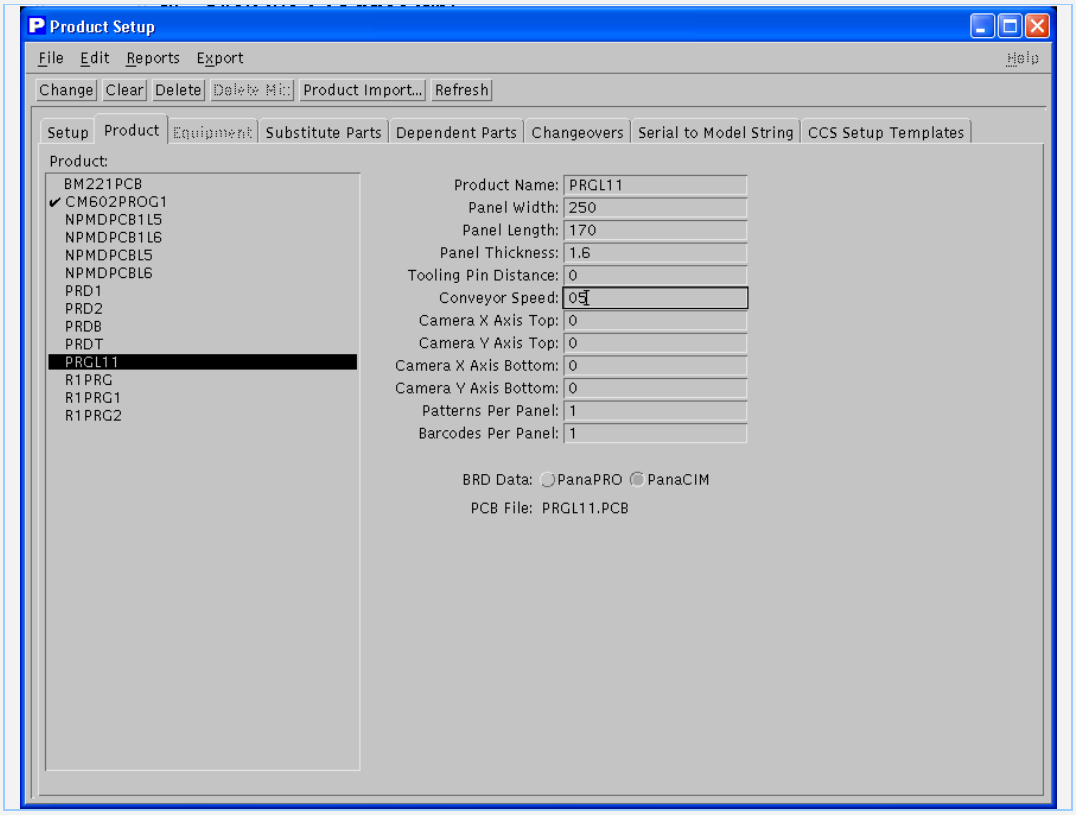
Step	Action
1	Open the **Product Setup** Application from the PanaCIM Main Menu.
2	Perform the following: a. Click the **Setup** tab. b. Select the **Product** from the Product list. c. Select the **Route**. The product may be setup to run on different production Lines. d. Select the **Mix** (or project). The same product may be in different mixes (combined setups). e. Enter the **Model String**. Product Changeover and Control requires that every product includes a unique Model Code (string) assigned that uniquely defines the product. Comparisons of model codes are made between approaching panels and panels currently running on the machine. If there is a difference between the two, a changeover is initiated. Model Codes and serial numbers are supplied by the STrack Scanner. Keep in mind: • STrack Scanners can be positioned above the panel, below the panel, or both. • You must specify whether the barcode is on the top or bottom side of the panel. f. Enter the **PT200 Group** and **PT200 Lot** (*for CM/DT series machines only*). g. Choose **Enable** if Substitute parts are allowed to be used in the product. *Refer to the Material Verification Manual for more information on using Substitute parts.* h. Enter **Top** or **Bottom** to indicate the side the barcode is located.



3

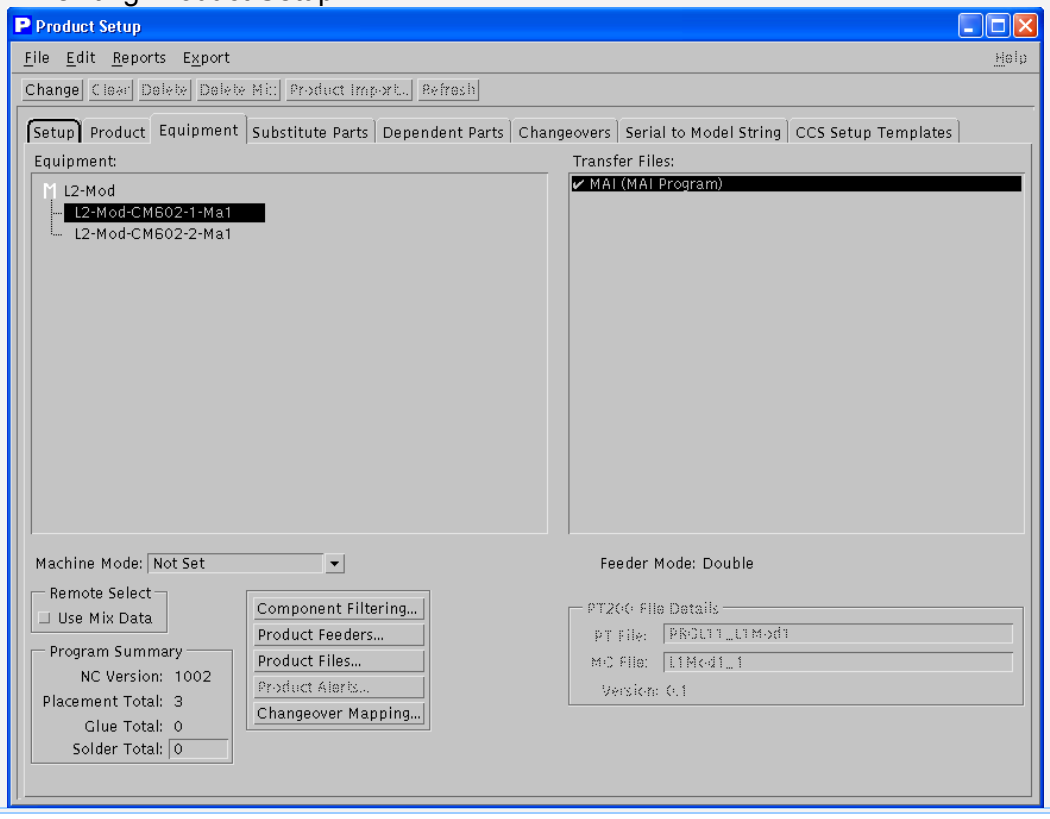
Perform the following:

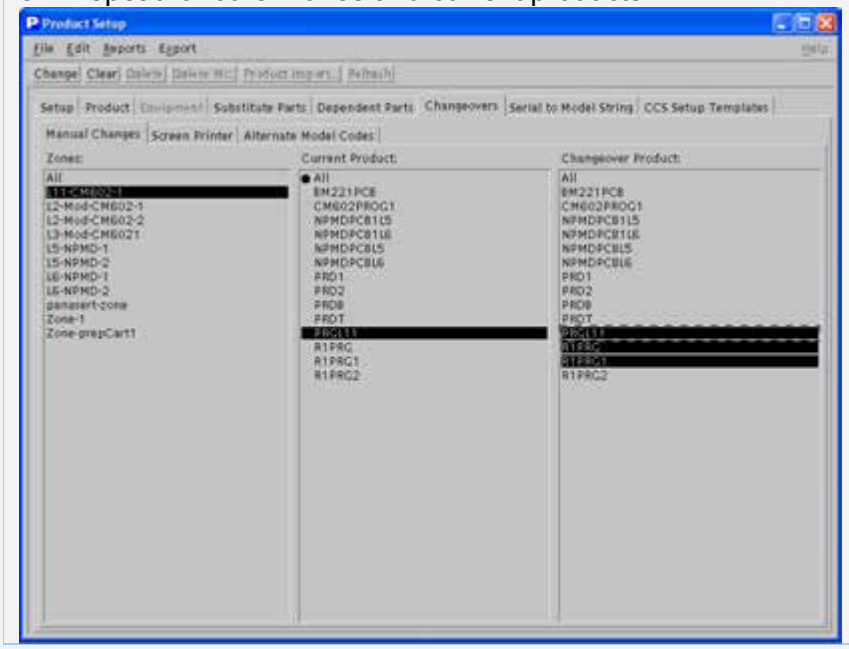
a. Click the **Product** tab.
b. Select the **Product** from the list.
c. **BRD** Data:
   - Select **PanaPRO** if the product was downloaded from the PanaPRO or DGS system to PanaCIM. In this case, all the other information is supplied in the Product.
   - Select **PanaCIM** if the product originates from some other source than PanaPRO or DGS. In this case the user needs to enter the rest of the information.



4 Perform the following:

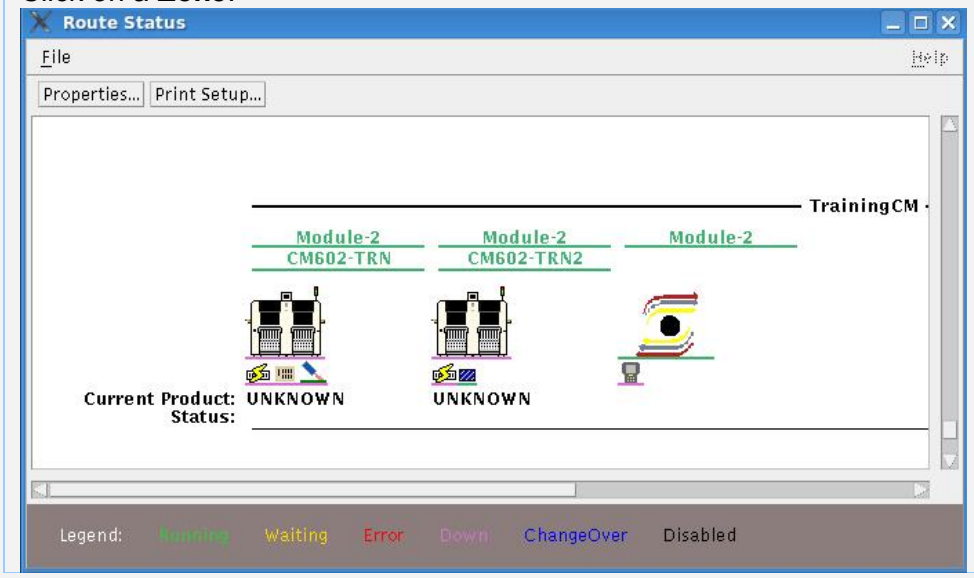
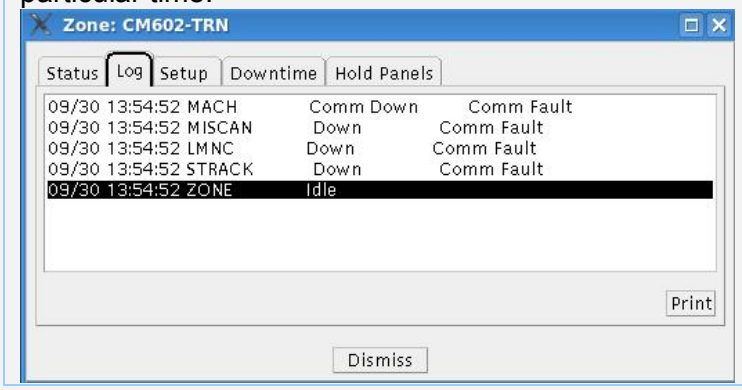
a. Click the **Equipment** tab.
b. Select the **Machine Name** from the Equipment area.
c. Select the files to download from PanaCIM to the PT200 or Machine from the **Transfer Files** area.
d. Click **Change** after all changes have been made to the product before exiting Product Setup.

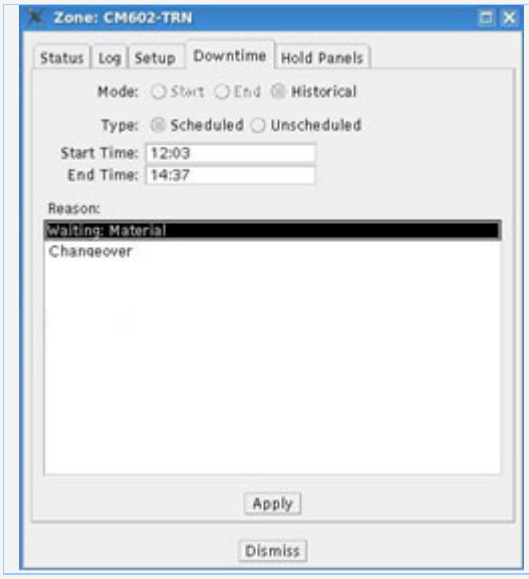


5	Select the **Changeovers** tab if **Line Managed** mode is active. Here you will indicate the new product changeovers that require no manual operation when changing from the current product. a. Select the **zone(s)** from the list. Select multiple zones by pressing **[Shift]** or **[Ctrl]**. b. Select a Product from the **Current Product** list. c. Select **Changeover Product(s)** that require no manual operation. d. Click **Change**. e. Repeat for other zones and current products. 

■ Route Status Operations

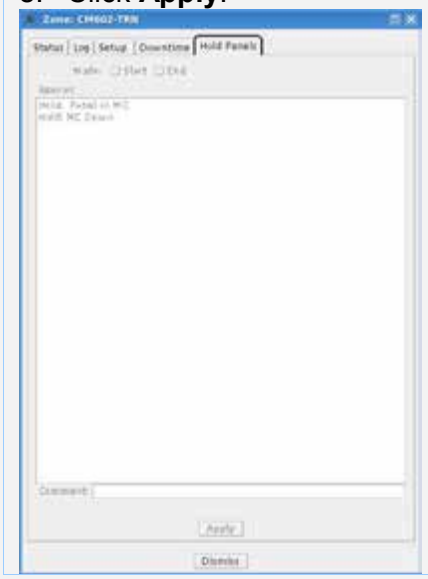
Refer to the action table below to perform various actions in Work Zones during Product Changeover and Control operations.

Step	Action
1	Open the **Route Status** Application from the PanaCIM Main Menu.
2	Click on a **Zone**. 
3	Select the **Log** tab to determine what problems or events have occurred at a particular time. 
4	Select **Downtime** to set machine downtimes. **Note**: Downtime reasons can be modified in the System Configuration Application. a. Click the **Downtime** tab. b. Mode: • Select **Start** to indicate the beginning of a machine down condition. • Select **End** to indicate the End of a machine down condition. • Select **Historical** to indicate a range of time in which the Zone is down. Enter **Start** and **End** times. c. Choose **Scheduled** or **Unscheduled**. A list of reasons will appear for your selection. d. Choose the **Reason** from the list. e. Click **Apply**.



5 Operators can hold the panel at the entrance of the Work Zone using the **Hold Panels** tab in the Route Status Application. Use Route Status to indicate a reason for the hold (either at the beginning or the end of the hold).

a. Select the **Hold Panels** tab.
b. Select **Start** or **End** to indicate when the reason should be chosen.
c. Select a **Reason** from the list for stopping production.
d. Add a **Comment** if desired.
e. Click **Apply**.



3.3 Operation

Refer to the action table below for an overview of the Changeover Monitoring operation.

Step	Action
1	The Panel enters the Production Zone.
2	The scanner will acquire the Model string of the panel. **Note:** If the device is unable to read a valid Model string, a Failed Read Error occurs.
3	The Model String is checked to see if it is mapped to a valid product. **Note:** If the Model string is not mapped to a valid product, an Invalid Model Code Error results.
4	PanaCIM compares the current product Model Code with the scanned Model Code. After a successful match, the panel is then released into the Production Zone and is processed by the Work Zone. The Product Changeover and Control process awaits the next panel. **Note:** If the codes do not match, the panel is stopped by the conveyor interlock. If PanaCIM is configured for: • **Automatic Changeover:** The process continues as Automatic Changeover based on the style of Machines in the Work Zone (see Automatic Changeover section). • **Automatic Detection:** The LMNC display prompts the operator to perform Manual Changeover. Once the operator performs the changeover, it displays on the LMNC.

3.3.1 Changeover Monitoring Operation

■CM/DT Series

This operation occurs when:

- Product Changeover and Control is set for Automatic Changeover Mode.
- The Panel Model Code does not match the current Production Model Code.

Refer to the action table below for a description of Automatic Changeover Operations for the CM/DT series machines.

Step	Action
1	The LMNC will display “Automatic Changeover in Progress”.
2	If the machines in the Work Zone are in Production mode: a. PanaCIM sends a remote stop command to each machine in the Work Zone. b. When each machine completes the remaining panels, it returns to the Main Menu on the control. c. All production data for the current product is uploaded to PanaCIM. d. PanaCIM activates the new program on each machine in the Work Zone. e. PanaCIM sends the remote start command to each machine in the Work Zone.
3	Each machine starts production and checks the product feeder setup. If the feeder and nozzle setup is *not correct*, PanaCIM will receive a machine event stating the start was not successful. You should correct any problems and then manually initiate production on the machine.
4	When PanaCIM receives a successful event from each machine in Work Zone, the conveyor interlock is released, and the panel is released into the Work Zone.

4. FUNCTION - PRODUCT CHANGEOVER WITH THE ASSOCIATED WORK ORDER

This chapter explains the basic operation of Product Changeover (Work Order).

4.1 Overview

In CIMC, functionality has been added that facilitates initiating changeover from the mobile scanner for model string and its associated work order. To initiate product changeover with the associated work order, model string and work order number can be entered / scanned on the mobile scanner. Basically, after scanning/entering the model barcode using the handheld scanner, PCC (product changeover and control module) validates whether it exists in the PanaCIM database. Only upon successful validation of model barcode, operator is allowed to proceed further on the scanner. Once the changeover procedure is completed on the scanner, program is downloaded to the machine.

4.2 Settings

4.2.1 Constraints

- Direction is host to equipment and selection is made on the machine in the case of CM/DT series machines.
- For CM/DT series modules, only single lane work order association is supported.
- Mobile scanner initiated changeover only applies to single lane CM/DT series modules.

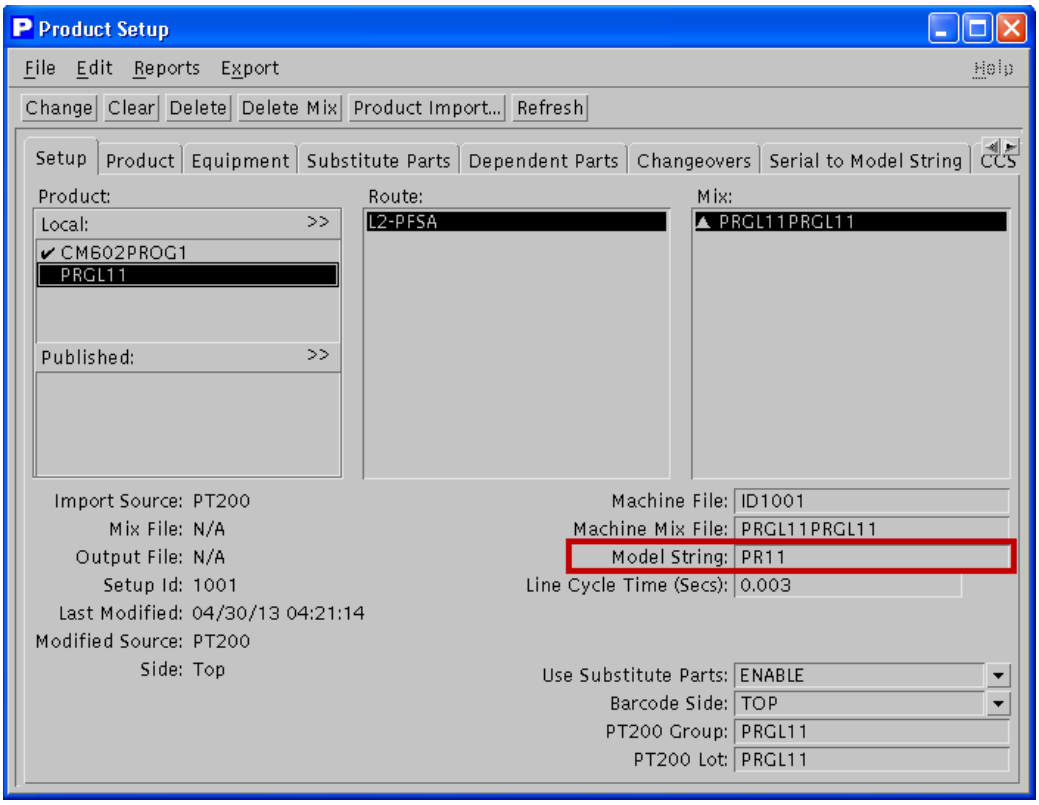
4.2.2 Licensing

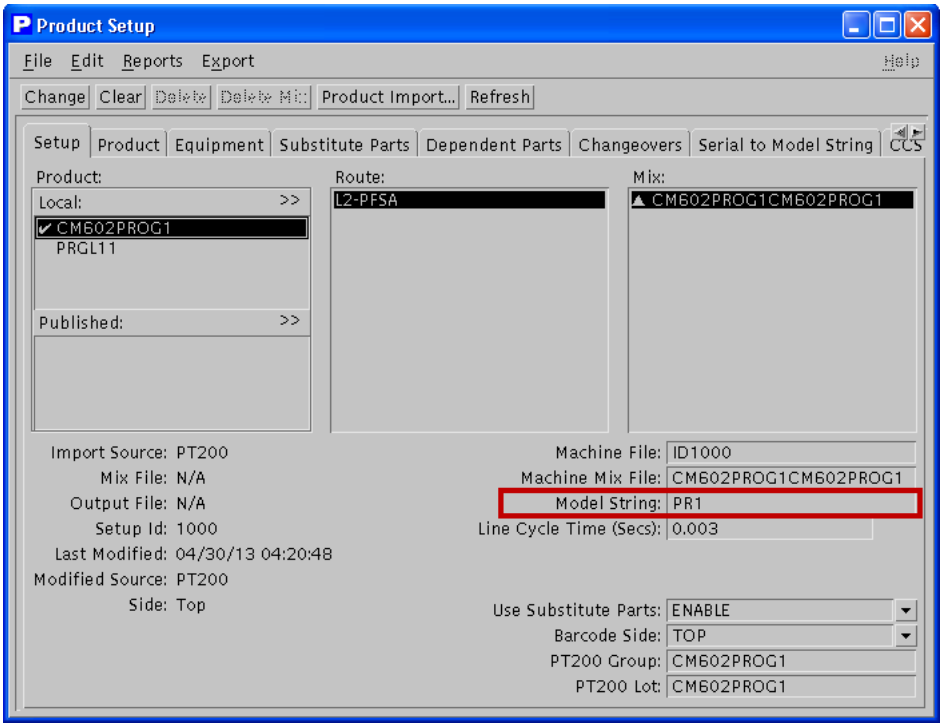
- MV (material verification). PCC (Product Changeover and Control) license expected in the case of changeover via the LMNC.

4.3 Operation

4.3.1 Initiating Changeover on the Scanner

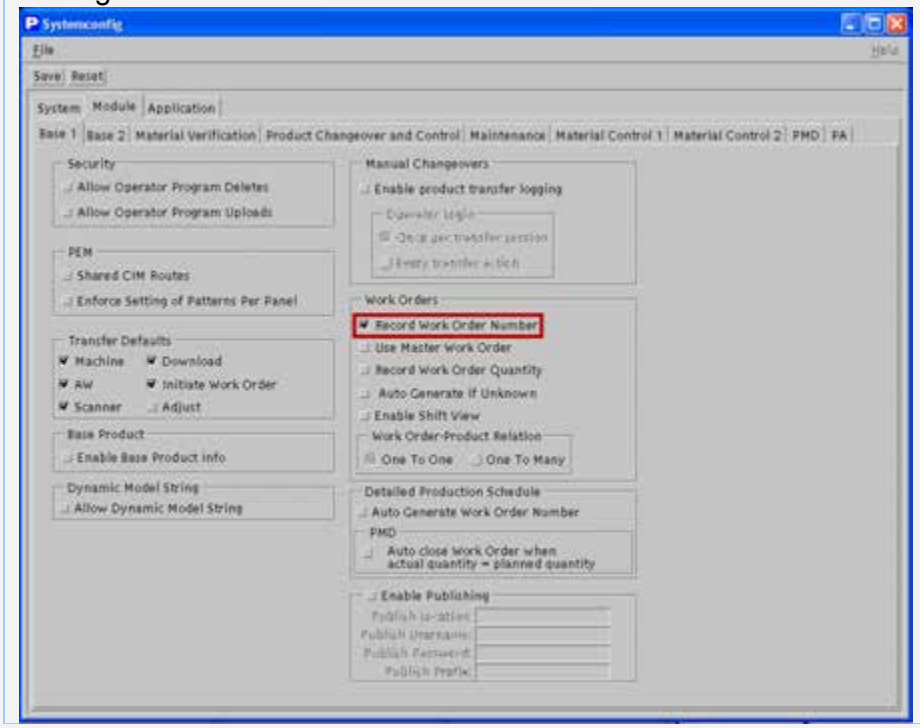
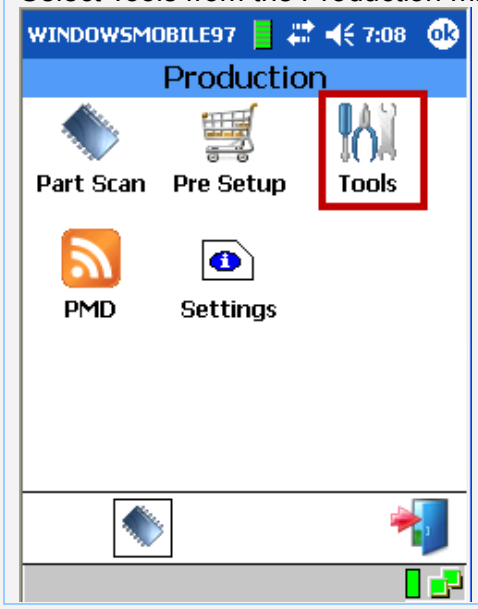
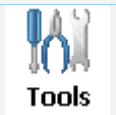
Before initiating changeover on the scanner, model string is defined in the product setup application as shown below:

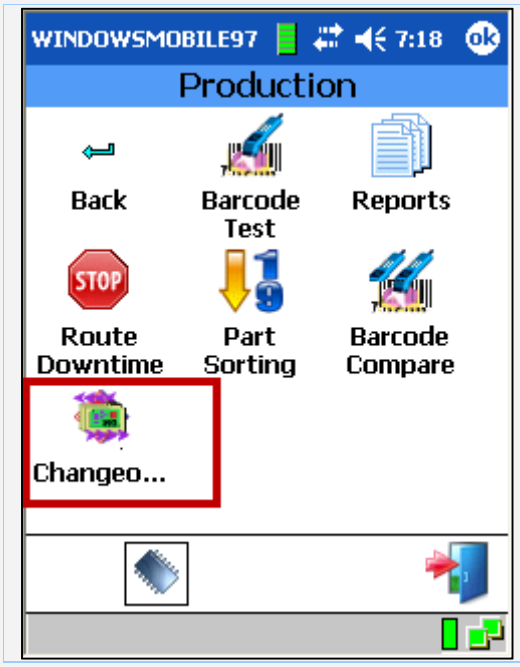
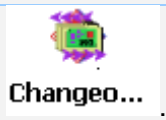
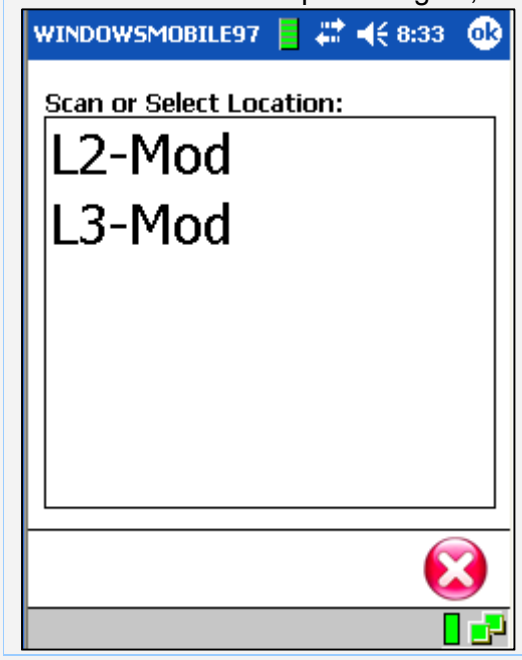


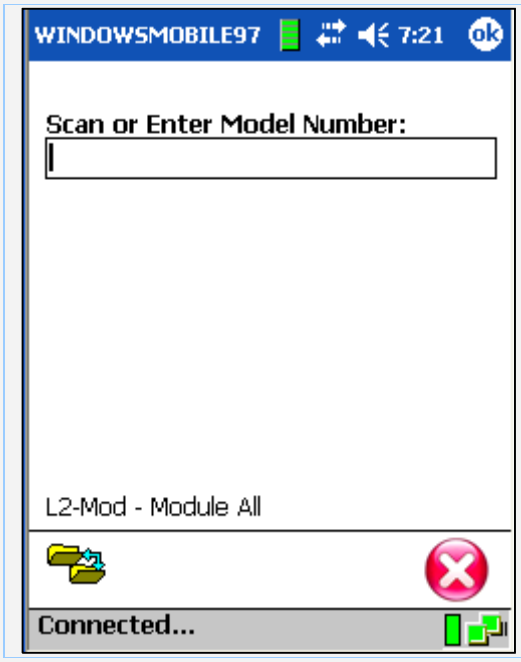


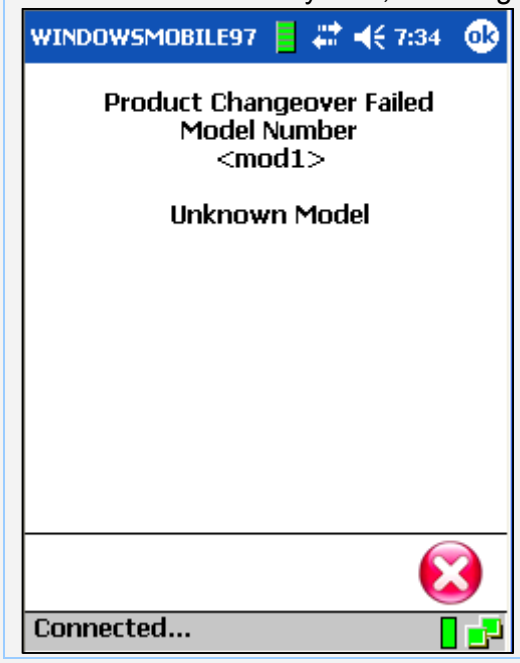
Refer to the action table below for the screens observed in the changeover when work order related

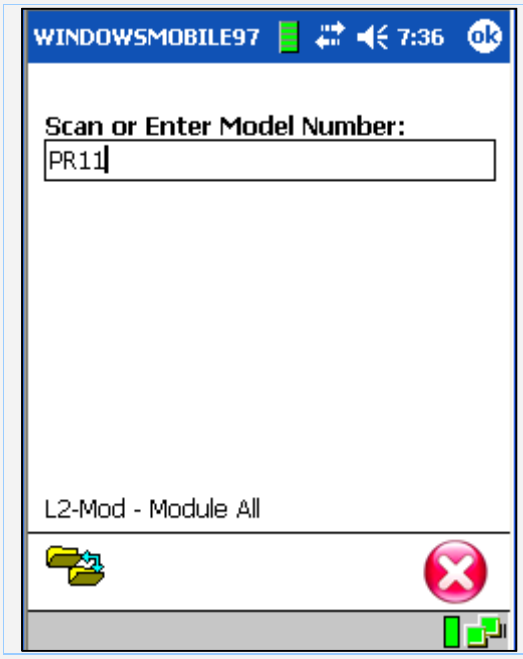
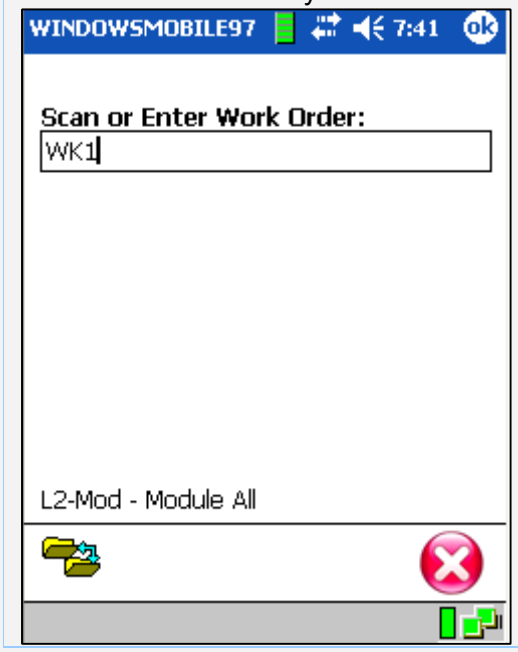
settings are activated in the system config application.

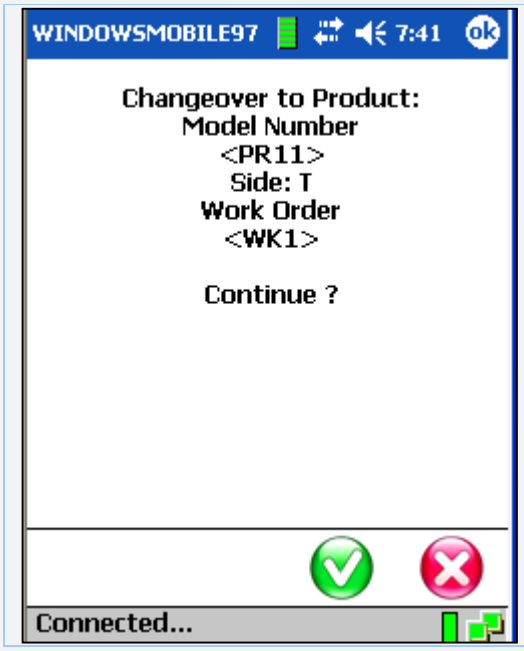
Step	Action
1	Settings **Record Work Order Number** are enabled as shown below: 
2	Select Tools from the Production main menu as shown below: 
3	Upon selecting  , following screen appears:

	
4	Click  . Upon doing so, following screen appears: 
5	Select the name of the module from the above shown screen. After doing so, following screen appears. As shown below, the module with **"Module All"** is selected automatically. User does not have option to select work area.



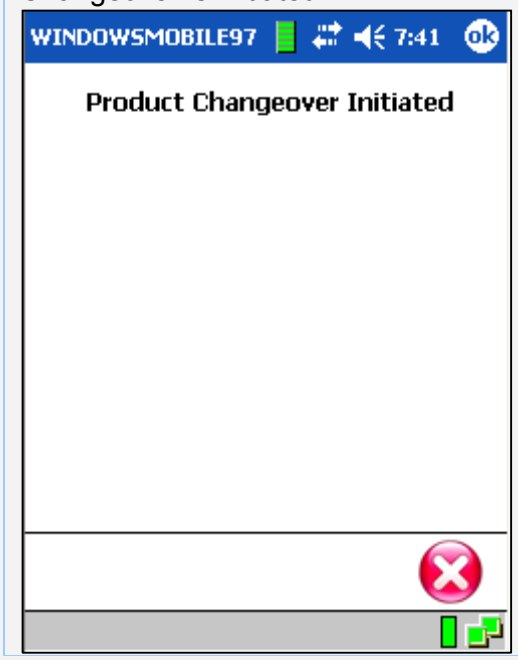
6	Scan or enter the model string defined in the product setup. If Model String does not exist in the system, following message is displayed: 
7	Scan or enter the model string existing in the system that needs to be activated as part of changeover.

	
8	After scanning/entering the model number, following screen appears to scan/enter work order. Work order name can identify a new work order that does not exist in the system or a work order existing in the system. 
9	After scanning/entering the work order, following message appears on the scanner prompting to changeover to product whose model number, side and work order are the ones input earlier (as shown in screens above).



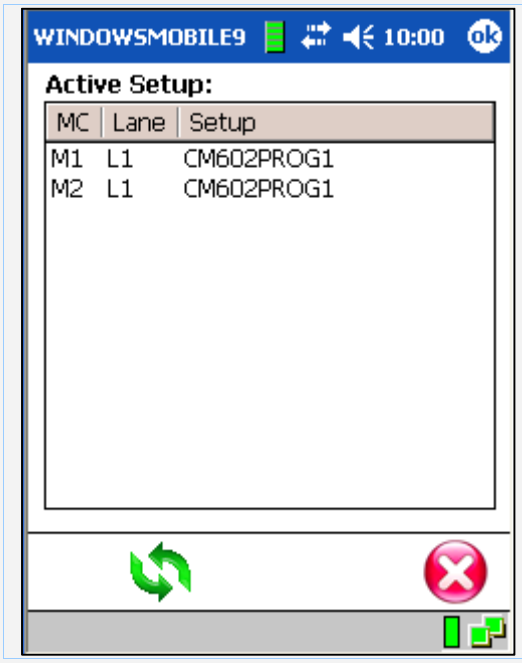
10

After clicking the  , following message appears indicating that Product Changeover is initiated.

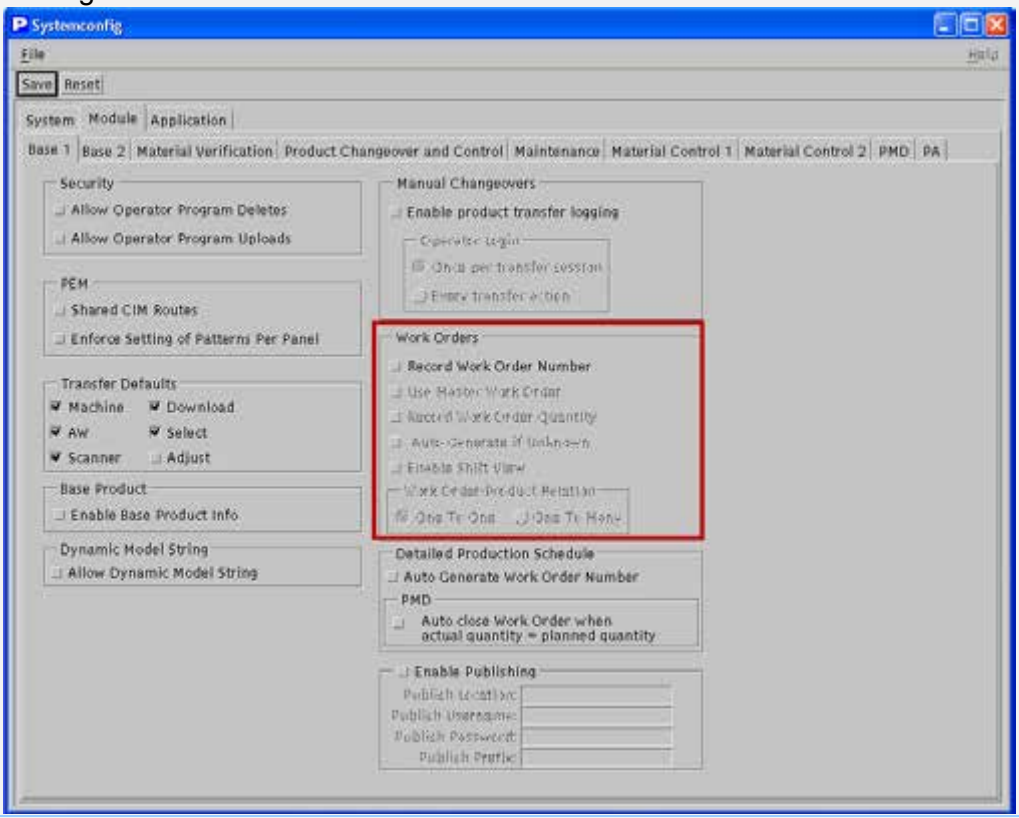


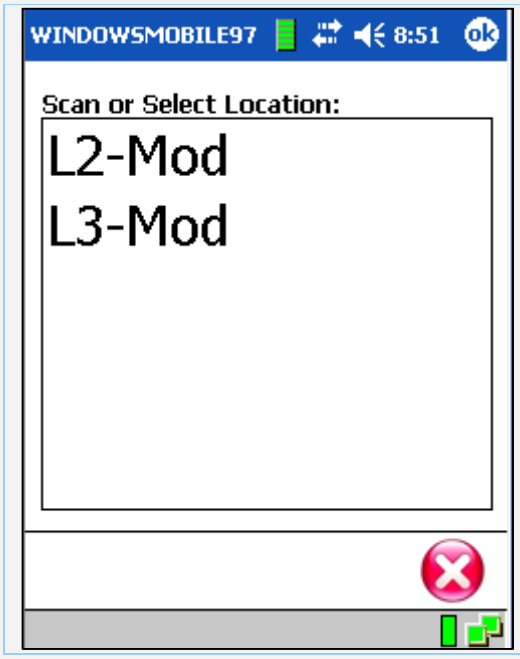
Note:

Clicking  displayed at the bottom while specifying model number / work order name shows the currently active setup.

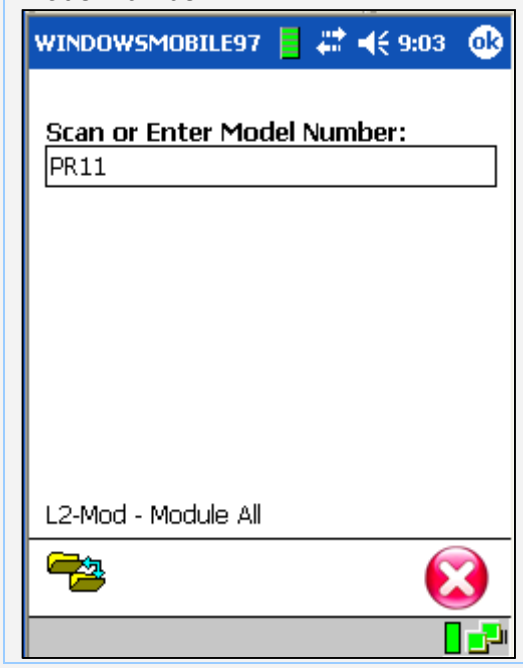


Refer to the action table below for the screens observed in the changeover when work order related settings are disabled in the system config application.

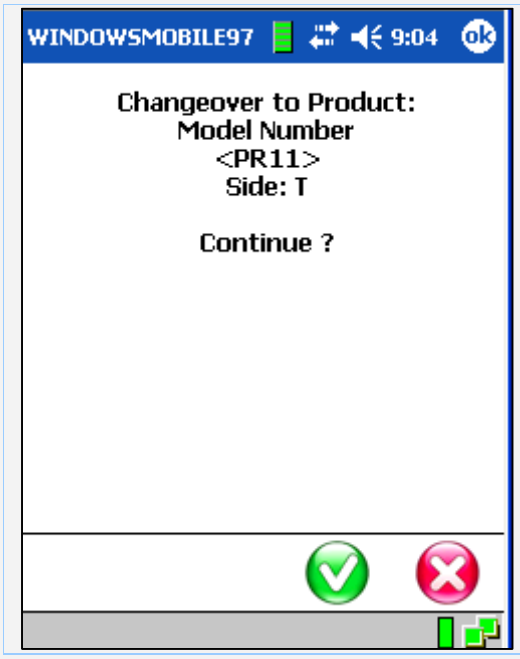
Step	Action
1	Settings **Record Work Order Number** are disabled as shown below: 
2	From  , select  . Upon doing so, following screen appears listing module names:



3	Select Module name. Upon doing so, following screen appears to scan/enter model number.

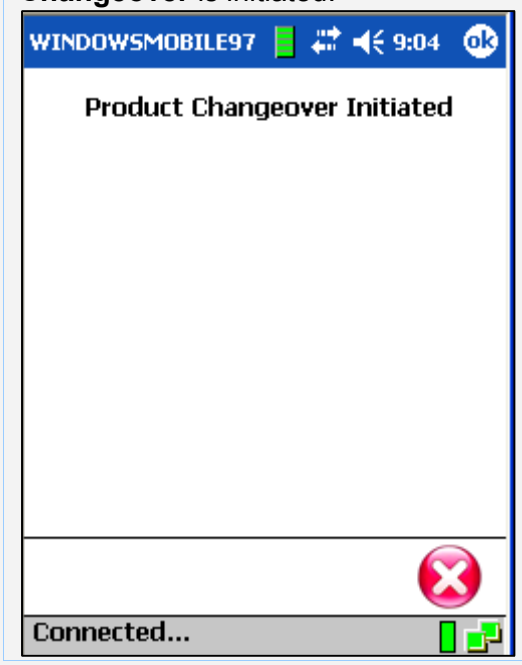


4	After model number is input, following screen appears prompting to changeover to product whose name is input above.



5

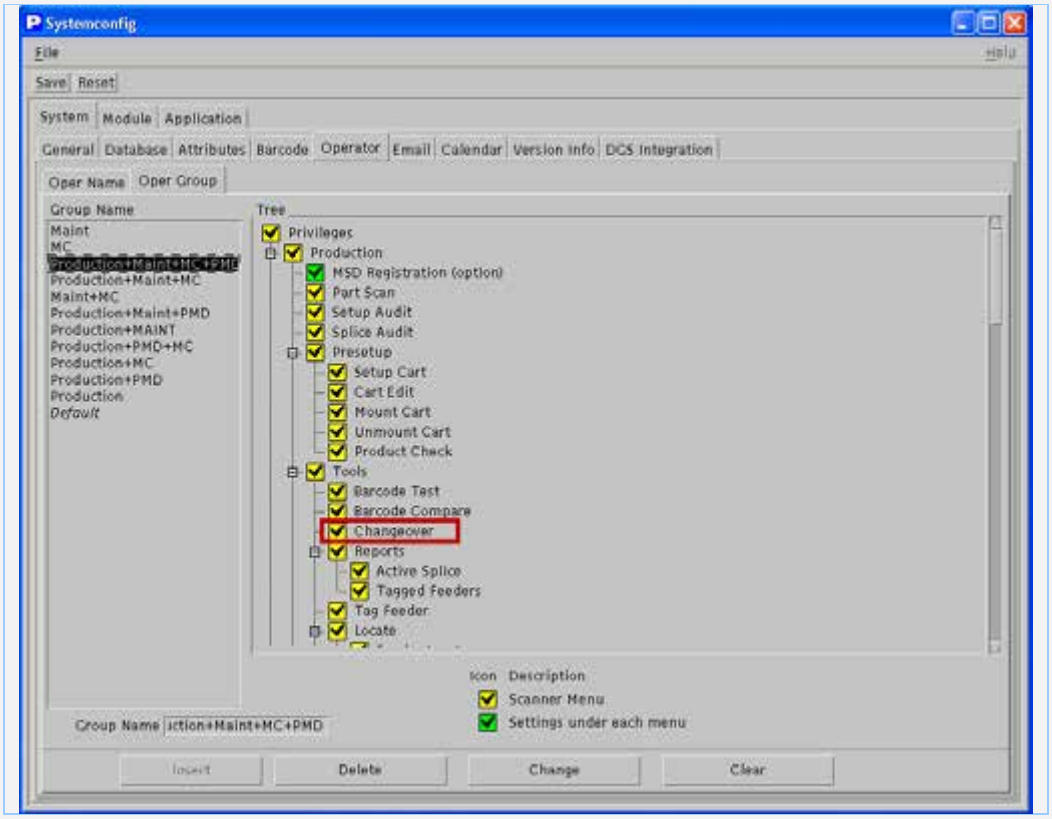
Click the  button. After doing so, a message appears indicating **Product Changeover** is initiated.



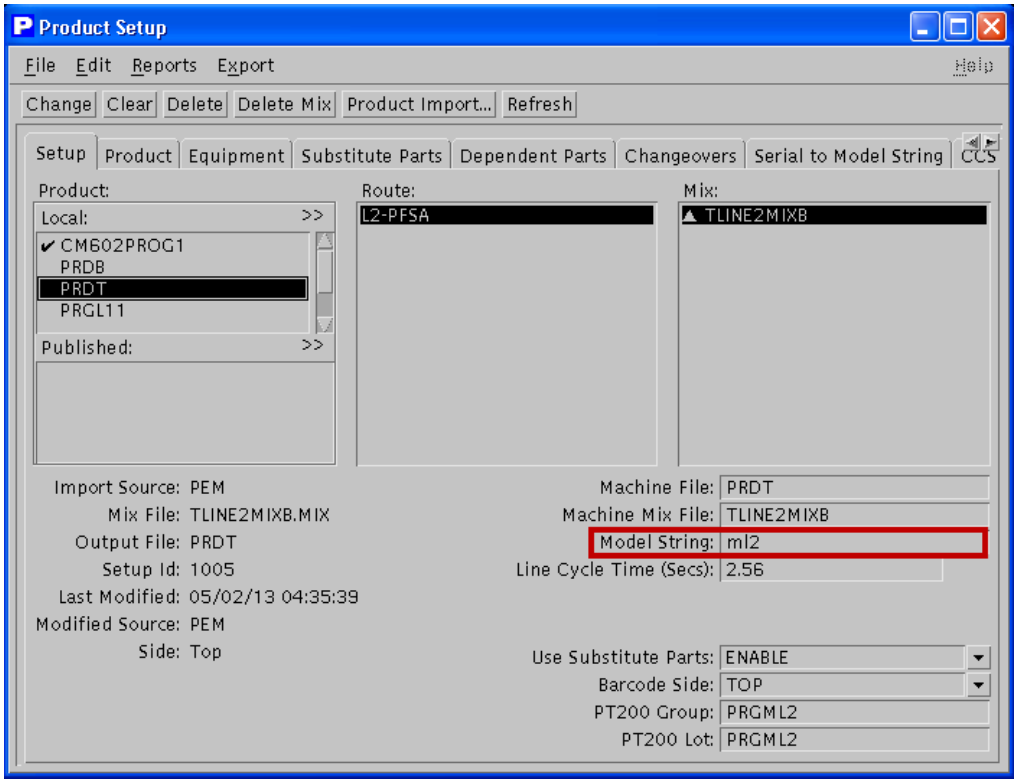
Note:

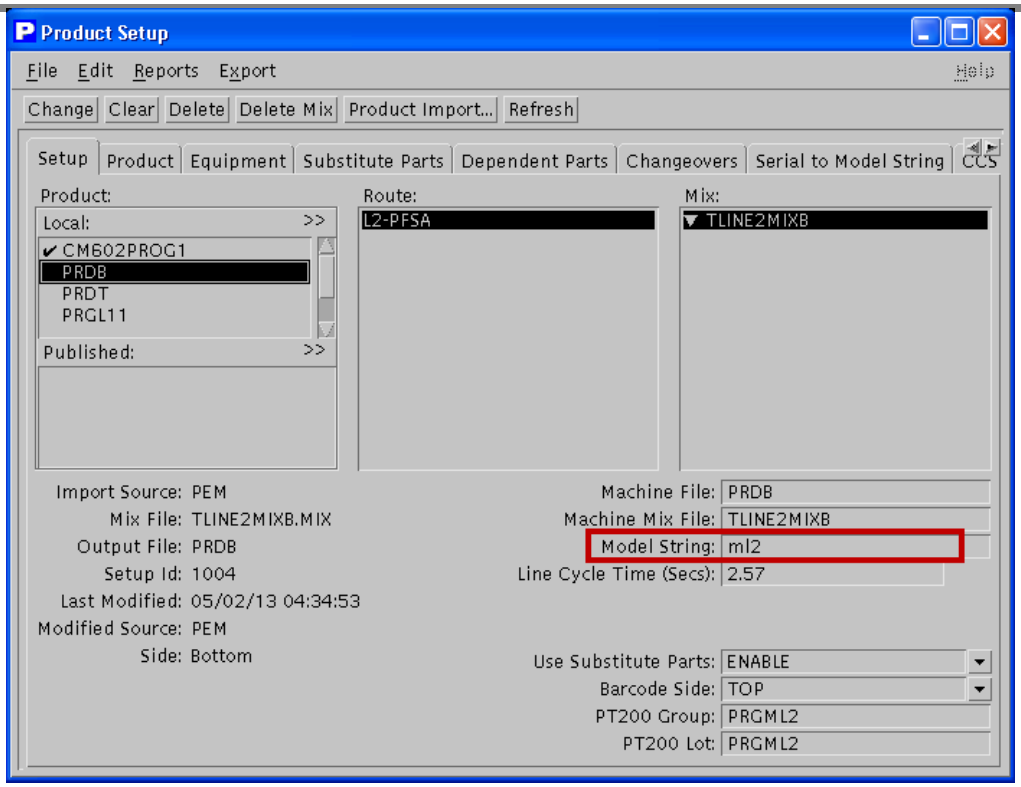
This 

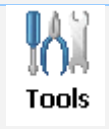
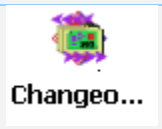
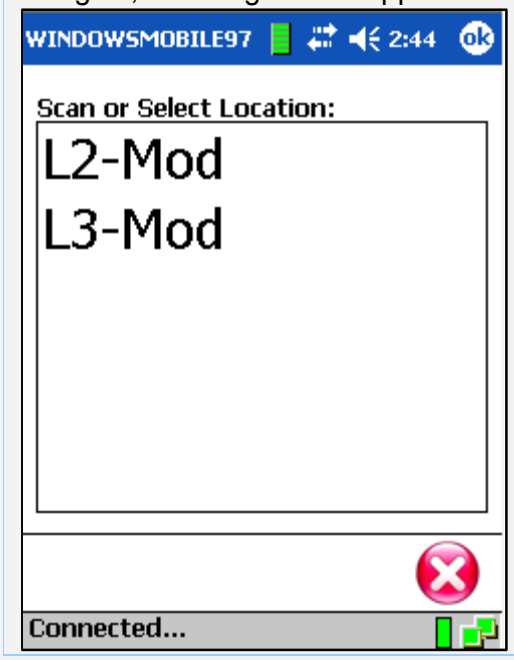
icon is available all the time in production mode only irrespective of license provided Changeover group privilege is enabled for the operator in the **Oper Group** tab.

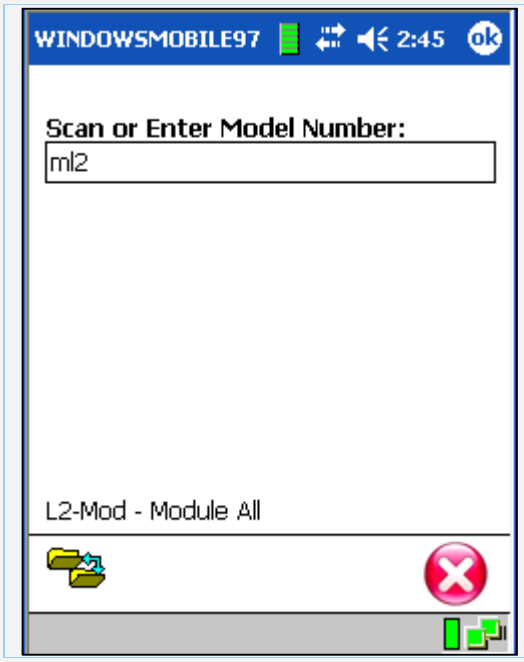


Refer to the screens below when model string contains both top and bottom side products.





Step	Action
1	Select  and then  from the production main menu. Upon doing so, following screen appears: 
2	Select the module name. Upon doing so, following screen appears:



3	Since the model string entered in the previous screen has both top and bottom side product associated, following screen appears prompting to select Top or Bottom side product.

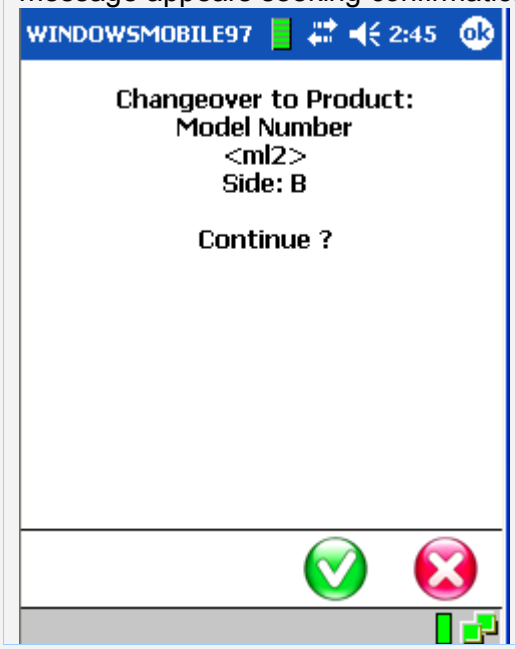


4	Shift focus on the Bottom button to select it.



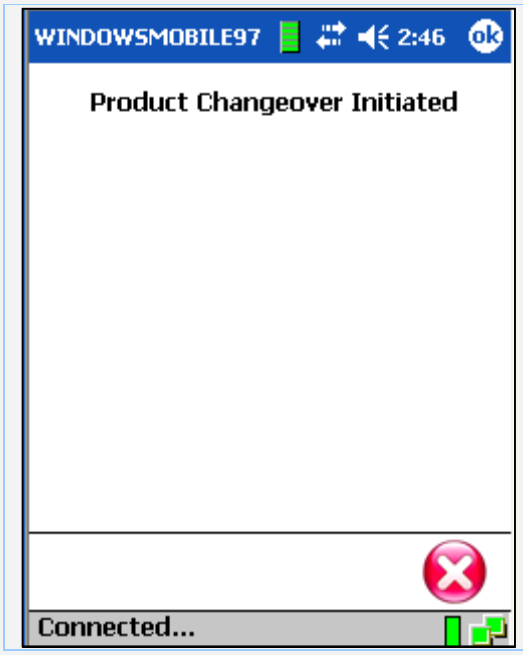
5

Click  after selecting top or bottom side product. Upon doing so, a message appears seeking confirmation to changeover to selected side.

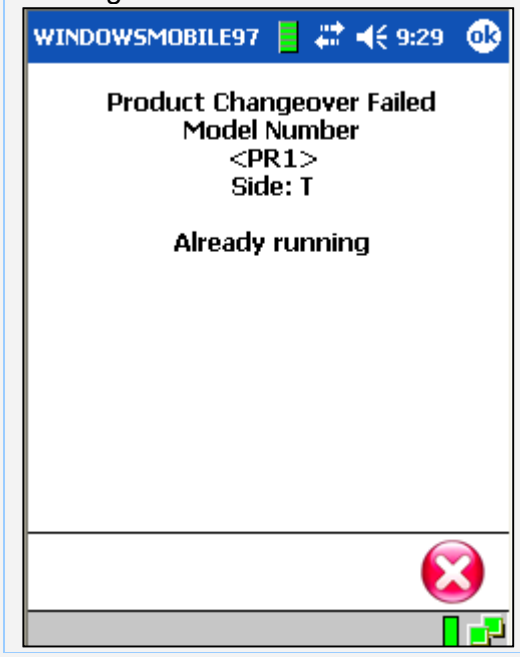


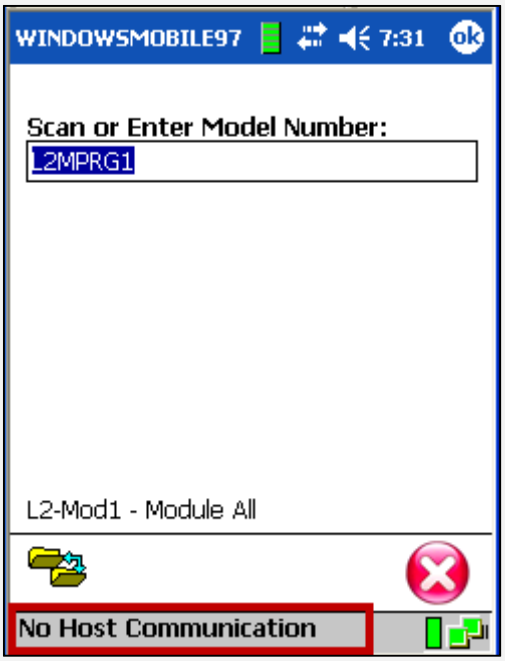
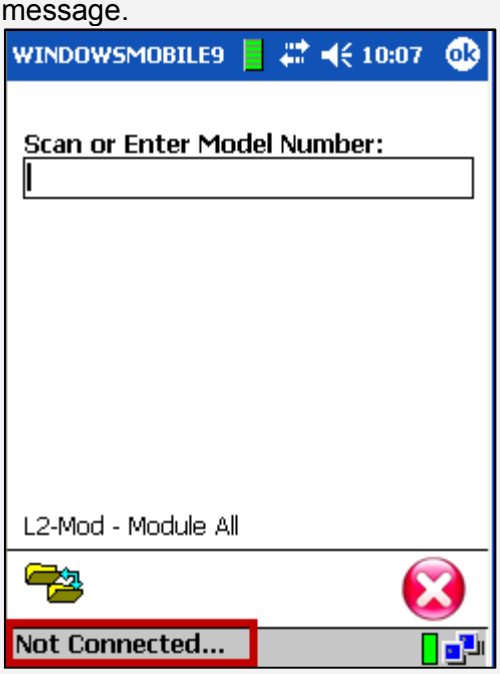
6

Click  . Upon doing so, product changeover is initiated.


WINDOWSMOBILE97
2:46
ok
Product Changeover Initiated
Connected...

4.3.2 Additional Messages on the Scanner

Message #	Description
1	Changing over to product that is already running displays the following message. 
2	If Product changeover license is unavailable, following message is displayed on the scanner upon trying to do changeover. 

3	This is the error which will appear anytime while communicating with PanaCIM host and host has not responded back even though MV license is enabled. 
4	Once MV license is cleared, Module Interface does not come-up and scanner can't communicate with host. Scanner shows **Not Connected** message. 

5. TROUBLESHOOTING

This chapter explains troubleshooting.

5.1 Overview

Most errors are displayed on the LMNC. *Refer to the Line Manager Node Controller document for more detailed troubleshooting information*.

5.2 Troubleshooting

Refer to the table below for a description of common errors displayed on the LMNC.

Error	Description	Remedy
Input Device Read Error	If an Input Device experiences an error acquiring the product identification, you are informed via an alarm function from the system's user interface.	Clear the alarm and rectify the error.
Incorrect Program (Recipe) Selection	An incorrect program or recipe was selected after a changeover is completed.	You will be alerted to change back to the correct program
Unregistered Product	The product indicated by the Model Code is not registered in PanaCIM.	Ensure the indicated product is registered in PanaCIM.
Communication Errors	The connection to one of the machines in the Work Zone or to PanaCIM has failed.	Ping to verify physical connections. Also verify that entries in the /etc/hosts file match Route Editor names.

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Printed in Japan